
Learning Informative Attention Weights for Person Re-Identification

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Abstract

Attention mechanisms are widely used in deep learning and have been incorporated into DNNs for person Re-Identification (Re-ID) to enhance discriminative feature learning. Existing attention modules, including self-attention and channel attention, learn attention weights that quantify the importance of feature tokens or feature channels. However, existing attention methods do not explicitly ensure that the attention weights are informative for predicting the identity of the person in the input image, and may consequently introduce noisy information from the input image. To address this issue, we propose a novel method termed Reduction of Information Bottleneck loss (RIB), motivated by the principle of the Information Bottleneck (IB). A novel distribution-free and efficient variational upper bound for the IB loss (IBB), which can be optimized by standard SGD, is derived and incorporated into the training loss of the RIB models. RIB is applied to DNNs with self-attention through a novel Differentiable Channel Selection Attention module (DCS-Attention), which selects informative channels to compute attention weights, leading to RIB-DCS. It is also integrated with existing channel attention modules to learn informative channel weights, leading to RIB-CA. Both RIB-DCS and RIB-CA are applied to fixed backbones and learnable backbones with Differentiable Neural Architecture Search (DNAS). Extensive experiments on multiple person Re-ID benchmarks show that RIB significantly improves prediction accuracy, including in occluded person Re-ID, by learning more discriminative and identity-informative features. The code of our work is available at <https://github.com/Statistical-Deep-Learning/RIB-ReID>.

1 INTRODUCTION

Attention methods, including self-attention and channel attention, have recently gained significant interest in computer vision tasks such as person re-identification (Re-ID). Existing self-attention modules [Wang et al., 2018, Zhang et al., 2020, Chen et al., 2019, He et al., 2021] compute attention weights by modeling pairwise affinities among feature tokens, while channel attention modules assign adaptive weights to feature channels [Hu et al., 2018, Woo et al., 2018, Jiang et al., 2024]. In both cases, accurate computation of attention weights is critical. However, these methods do not explicitly ensure that attention weights are informative for predicting the class label, i.e., person identity in person Re-ID. This issue is illustrated by the Grad-CAM visualization of the attention-based vision transformer SPT [Tan et al., 2024] in Fig. 1(a)-(b), where the model focuses on occlusion regions unrelated to identity in the occluded person Re-ID task. For another example, Fig. 1(c)-(d) shows that feature tokens from occluding regions receive high attention weights for a query token from the boundary of the arm. The Grad-CAM visualization in Fig. 3(c) in Section A.15 further shows that representative baselines in self-attention, TransReID [He et al., 2021], and channel attention, SE [Hu et al., 2018], either focus on background areas or miss important human body regions, potentially degrading person Re-ID performance. These examples indicate that existing attention-based methods lack a principled information-theoretic framework to ensure that higher attention weights are assigned to identity-relevant and informative features.

To address this issue, we propose to learn informative attention weights in existing self-attention and channel attention modules so that the resultant DNNs learn discriminative features for person Re-ID. By learning informative attention weights, DNNs assign higher attention to feature tokens in self-attention modules and to feature channels in channel attention modules that contribute more to the discriminative task of person Re-ID. Our proposed Reduction of Information Bottleneck loss (RIB) achieves this goal by reducing

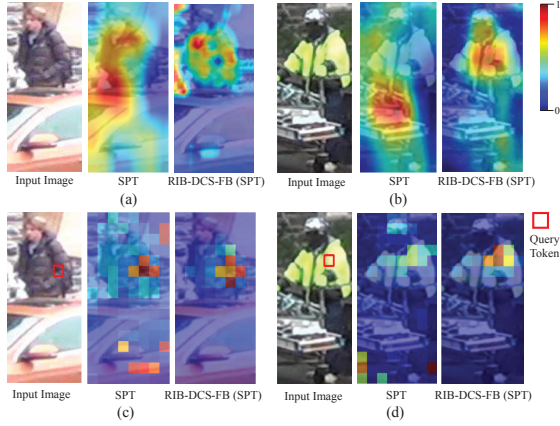


Figure 1: Figures (a)-(b) illustrate the Grad-CAM visualization for two images from the occluded person Re-ID dataset, Occluded-Duke [Miao et al., 2019], for an attention-based model, SPT [Tan et al., 2024], and RIB-DCS-FB (SPT). Figures (c)-(d) illustrate the heatmaps of the attention weights corresponding to a query token computed from the first transformer block in SPT and RIB-DCS-FB (SPT). The query token for both examples is selected from the boundary of the arm of the person in the images, which is critical for identifying a person. More visualization results are deferred to Section A.15 of the supplementary.

the Information Bottleneck (IB) loss, as detailed later in this section, inspired by the IB principle [Tishby et al., 1999]. RIB is applied to DNNs with self-attention modules through a novel Differentiable Channel Selection Attention module (DCS-Attention), which selects more informative channels for learning more informative attention weights than vanilla self-attention modules, leading to models termed RIB-DCS. RIB is also incorporated into DNNs with existing channel attention modules to learn more informative channel attention weights than existing channel attention modules, leading to RIB-CA. Both RIB-DCS and RIB-CA are integrated into Fixed Backbones (FB) and learnable backbones with Differentiable Neural Architecture Search (DNAS), resulting in RIB-DCS-FB, RIB-DCS-DNAS, RIB-CA-FB, and RIB-CA-DNAS. As illustrated in Fig. 1(a)-(b), RIB-DCS-FB (SPT) focuses on important human parts rather than occlusion regions compared to the baseline self-attention model, SPT. Moreover, Fig. 1(c)-(d) illustrates that RIB-DCS-FB (SPT) assigns higher attention weights to informative and semantically relevant feature tokens from the boundary of the arm for a query from a similar location.

Motivation of RIB for Self-Attention and Channel Attention. Self-attention [He et al., 2021, Zhang et al., 2020] is formulated as $\text{Output} = \sigma(QK^\top)V$, where $Q, K, V \in \mathbb{R}^{N \times D}$ denote the query, key, and value, with N feature tokens and input channel number D . $\sigma(\cdot)$ is an operator such as Softmax. The attention weights $A = \sigma(QK^\top) \in \mathbb{R}^{N \times N}$

model the weighted aggregation of feature tokens in V . By learning informative A , DNNs assign higher weights to tokens in V that are more informative and semantically relevant to the person identity. For example, tokens corresponding to human body regions in Fig. 1(c)-(d) should receive higher weights than those from occlusions or background. We propose Differentiable Channel Selection Attention (DCS-Attention), which computes more informative A than vanilla self-attention by using only selected informative channels, i.e., selected columns of Q and corresponding rows of $K^\top$, to compute $A = \sigma(QK^\top)$, as shown in Fig. 2(a). Learning informative A for discriminative features can be interpreted via the Information Bottleneck (IB) principle. Let X be input features, F the learned features, and Y the ground truth class labels defining person identity in Re-ID. IB aims to increase $I(F, Y)$ while reducing $I(F, X)$, i.e., reducing the IB loss $I(F, X) - I(F, Y)$, where $I(\cdot, \cdot)$ denotes mutual information. By reducing the IB loss with RIB, DNNs are encouraged to select more informative channels for computing attention weights in self-attention, so that the attention output aggregates feature tokens in V more correlated with person identity and less correlated with background or occlusions. Table 7 in Section A.4 shows that replacing standard self-attention with DCS-Attention already reduces the IB loss and improves person Re-ID performance without explicitly optimizing the IB loss. Further explicitly reducing the IB loss with RIB leads to additional performance gains and an even lower IB loss, demonstrating the effectiveness of RIB. The difference and advantage of our hard channel selection in DCS-Attention over SE [Hu et al., 2018], CBAM [Woo et al., 2018], and MCA [Jiang et al., 2024] are detailed in Section E of the supplementary.

Similar to self-attention, existing channel attention modules compute weights from heuristic gating functions [Hu et al., 2018, Woo et al., 2018, Jiang et al., 2024], without explicitly enforcing that higher weights correspond to more informative channels for person Re-ID. Under the RIB framework, RIB-CA enhances SE [Hu et al., 2018], CBAM [Woo et al., 2018], and MCA [Jiang et al., 2024] by assigning higher weights to more discriminative feature channels. RIB is especially effective for occluded person Re-ID. By reducing the correlation between learned features and occluded inputs while increasing their correlation with class labels, the network learns features that are robust to occlusions, as shown in Fig. 1 and Table 4.

Contributions. First, we propose a novel distribution-free and efficient Information Bottleneck reduction framework, termed RIB, which learns informative attention weights for self-attention and channel attention modules by explicitly reducing the IB loss during training of DNNs with attention modules for person Re-ID. To reduce the IB loss, we derive a novel and provable variational upper bound, termed IBB, which can be optimized via standard SGD. The IB loss is reduced by reducing its upper bound, IBB. Unlike

existing bounds such as VIB [Dai et al., 2018, Srivastava et al., 2021], which assume Gaussian hidden features, and APIB [Guo et al., 2023], which optimizes only an approximation to the IB loss, RIB directly reduces the variational upper bound IBB without any distributional assumptions on hidden features. Moreover, IBB is computationally efficient with complexity $\Theta(nCT_0)$, where C is the number of classes, n the number of training samples, and T_0 the cost of a forward and backward pass per sample. In contrast, CLUB [Cheng et al., 2020], though distribution-free, requires $\Theta(n^2T_0)$ since $n \gg C$. A composite loss combining IBB with cross-entropy and triplet losses is used to train DNNs with attention modules for person Re-ID. Table 7 in Section A.4 shows that the IB loss is effectively reduced by this composite loss, and Table 3 in Section 4.4 shows that IBB outperforms CLUB, VIB, and APIB. As an independent contribution, the RIB framework is applicable to broader discriminative tasks beyond person Re-ID.

Second, RIB is applied to DNNs with self-attention or channel attention modules to learn informative attention weights. Under this framework, we propose DCS-Attention, which selects more informative channels to compute informative self-attention weights, leading to RIB-DCS. RIB is also integrated with existing channel attention modules, including SE [Hu et al., 2018], CBAM [Woo et al., 2018], and MCA [Jiang et al., 2024], to learn informative channel attention weights, leading to RIB-CA. Both RIB-DCS and RIB-CA are applied to Fixed Backbones (FB) and learnable backbones with Differentiable Neural Architecture Search (DNAS). Experiments in Section 4.2 and Section 4.3 show that they significantly outperform baselines without RIB on public person Re-ID benchmarks. Visualization results in Fig. 1, Fig. 3, and Fig. 4-7 in Section A.15 of the supplementary illustrate the advantages of RIB compared to existing self-attention and channel attention modules. RIB models also achieve superior performance on challenging settings, including occluded, cross-domain, and self-supervised person Re-ID, as shown in Section 4.5 and Section A.12-A.13.

2 RELATED WORKS

Related Works about Person Re-ID and Attention Methods. Person Re-ID aims to match individuals across different camera views. Both supervised [Quan et al., 2019, Zhang et al., 2023, He et al., 2021] and unsupervised [Chen et al., 2024, Yang et al., 2024a] approaches have been extensively studied using DNNs. Motivated by the success of self-attention in vision [Woo et al., 2018, Zhao et al., 2020, Vaswani et al., 2017, Wang et al., 2018], prior works incorporate self-attention or transformer modules for capturing global dependencies [Zhang et al., 2020, Chen et al., 2019, Quan et al., 2019, Zhang et al., 2023, He et al., 2021]. Some methods additionally exploit external meta information [Zhao et al., 2017, Zheng et al., 2019, Qin et al., 2024,

Yang et al., 2024b, Guo et al., 2024, Huang et al., 2024], which are complementary to our approach. Beyond self-attention, channel attention mechanisms [Hu et al., 2018, Woo et al., 2018, Jiang et al., 2024] enhance feature representation by adaptively re-weighting feature channels.

Neural Architecture Search Neural Architecture Search (NAS) methods are broadly categorized into differentiable NAS (DNAS) and non-differentiable NAS. Non-differentiable approaches rely on reinforcement learning [Zoph and Le, 2017] or evolutionary algorithms [Real et al., 2019] to explore architectures. In contrast, DNAS methods [Liu et al., 2019] relax the discrete search space for gradient-based optimization over a predefined set of operations. For instance, FBNetV2 [Wu et al., 2019, Wan et al., 2020] employs Gumbel-Softmax masks to select filter numbers across convolutional layers.

Related Works about Information Bottleneck (IB) The IB principle [Tishby et al., 1999] aims to learn compact representations that retain task-relevant information while discarding input redundancy. Deep VIB [Alemi et al., 2017] formulates IB as a training objective for deep networks. Extending this idea, [Lai et al., 2021] design a spatial attention module to decrease mutual information with inputs and increase mutual information with labels, and [Zhou et al., 2022] interpret self-attention as recursive IB optimization. Recent analyses [Amjad and Geiger, 2020, Kawaguchi et al., 2023] further show that mitigating the information bottleneck can reduce generalization error in DNNs.

3 PROPOSED APPROACH

Attention-based DNNs, including self-attention and channel attention, have demonstrated strong performance for person Re-ID by dynamically learning discriminative features [Chen et al., 2019, Zhang et al., 2020, He et al., 2021]. In this section, we propose a novel, distribution-free, and efficient Information Bottleneck reduction framework, termed RIB, which learns informative attention weights for both self-attention and channel attention modules. Under the RIB framework, we introduce Differentiable Channel Selection Attention (DCS-Attention) to select more informative channels and produce more informative attention weights than vanilla self-attention. DNNs with DCS-Attention enhanced by RIB are termed RIB-DCS. RIB is also applied to existing channel attention modules to encourage higher weights on informative feature channels, and the resulting models are referred to as RIB-CA. In Section 3.1, we present the formulation of RIB and a novel distribution-free and efficient variational upper bound for the IB loss. In Section 3.2, we introduce how RIB is applied to DNNs with DCS-Attention and existing channel attention modules, which leads to RIB-DCS and RIB-CA. In Section 3.3, we introduce how RIB-DCS and RIB-CA can be employed in fixed backbones designed for person Re-ID and learnable backbones that can

be optimized by NAS methods.

3.1 RIB: REDUCTION OF INFORMATION BOTTLENECK LOSS

Although attention methods have achieved remarkable success in person Re-ID, existing self-attention and channel attention modules do not explicitly ensure that attention weights are informative for predicting person identity and may introduce information irrelevant to discriminative learning. To address this issue, we propose a novel information-theoretic feature learning method, termed Reduction of the Information Bottleneck loss (RIB), which encourages attention modules to learn informative attention weights by reducing the IB loss of the associated DNNs. We first introduce the IB loss and the motivation for reducing it, and then present a new distribution-free and efficient variational upper bound, which is optimized to reduce the IB loss.

RIB aims to reduce the IB loss, $\text{IB}(F, X, Y) = I(F, X) - I(F, Y)$, where $I(\cdot, \cdot)$ denotes mutual information. X , F , and Y represent the input feature, learned feature, and ground-truth class label. Reducing the IB loss encourages learned features to be more correlated with class labels (person identities) and less correlated with input features. Since existing attention modules do not explicitly enforce informative weighting, they may attend to less informative tokens or channels corresponding to backgrounds or occlusions. By reducing the IB loss, RIB promotes higher attention weights for informative feature tokens in self-attention and feature channels in channel attention by increasing correlation with class labels and decreasing correlation with input images, leading to more discriminative features. Given training data $\{X_i, y_i\}_{i=1}^n$, where X_i is the i -th input feature and y_i is its class label. Let $\text{IB}(\mathcal{W}) = I(F(\mathcal{W}), X) - I(F(\mathcal{W}), Y)$ denote the IB loss, where $\mathcal{W}$ are the DNN weights and X takes values in $\{X_i\}_{i=1}^n$. $F(\mathcal{W})$ takes values in $\{F_i(\mathcal{W})\}_{i=1}^n$ with $F_i(\mathcal{W})$ being the i -th learned feature by the network. $F(\mathcal{W})$ and $F_i(\mathcal{W})$ are also abbreviated as F and F_i for simplicity. Y takes values in $\{y_i\}_{i=1}^n$. We have the class centroids $\{\mathcal{C}_a\}_{a=1}^C$ and $\{\mathcal{C}_b\}_{b=1}^C$ for the learned features $\{F_i\}_{i=1}^n$ and the input features $\{X_i\}_{i=1}^n$, respectively, where each $\mathcal{C}_a$ is the average of the learned features in class a , each $\mathcal{C}_b$ is the average of the input features in class b , and C is the number of classes. Then we define the probability that F_i belongs to class $\mathcal{C}_a$ as $\Pr[F \in a] = \frac{1}{n} \sum_{i=1}^n \phi(F_i, a)$ with $\phi(F_i, a) = \frac{\exp(-\|F_i - \mathcal{C}_a\|_2^2)}{\sum_{a=1}^C \exp(-\|F_i - \mathcal{C}_a\|_2^2)}$. Similarly, we define the probability that X_i belongs to class $\mathcal{C}_b$ as $\Pr[X \in b] = \frac{1}{n} \sum_{i=1}^n \phi(X_i, b)$. The class-centroid soft assignment and the computation of the probability that F_i belongs to class a by $\Pr[F \in a] = \frac{1}{n} \sum_{i=1}^n \phi(F_i, a)$ follow the prototype-based feature distribution modeling in existing prototypical learning methods [Caron et al., 2020, Li et al., 2021a, Yang et al., 2018, Snell et al., 2017].

The success of prototype-based methods shows that neural networks are capable of producing latent features that can be well aligned with class/identity centroids. Moreover, the plug-in estimators $\Pr[F \in a]$, $\Pr[F \in a, Y = y]$, and $\Pr[F \in a, X \in b]$ are consistent estimators of the corresponding probabilities under standard regularity conditions, so that IBB is provably an upper bound for the true IB loss with consistency, as detailed in Section G of the supplementary. The class-centroid soft assignment is further extended to a general prototype-based soft assignment with more prototypes than classes in Section A.18 of the supplementary, leading to RIBPL. We can compute the mutual information $I(F, X)$ and $I(F, Y)$ by

$$I(F, X) = \sum_{a=1}^C \sum_{b=1}^C \Pr[F \in a, X \in b] \log \frac{\Pr[F \in a, X \in b]}{\Pr[F \in a] \Pr[X \in b]},$$

$$I(F, Y) = \sum_{a=1}^C \sum_{y=1}^C \Pr[F \in a, Y = y] \log \frac{\Pr[F \in a, Y = y]}{\Pr[F \in a] \Pr[Y = y]},$$

and then compute the IB loss $\text{IB}(\mathcal{W})$. Given a variational distribution $Q(F \in a | Y = y)$ for $y \in \{1, \dots, C\}$ and $a \in \{1, \dots, C\}$, the following theorem gives a variational upper bound, $\text{IBB}(\mathcal{W})$, for the IB loss $\text{IB}(\mathcal{W})$. $\text{IBB}(\mathcal{W})$ is also abbreviated as IBB in the following text.

Theorem 3.1. Let $\Pr[X \in y] = \sum_{i=1}^n \mathbb{I}_{\{y_i=y\}}/n := p_y$ be the prior probability for every $y \in [C]$, we have

$$\text{IB}(\mathcal{W}) \leq \text{IBB}(\mathcal{W}), \quad (1)$$

where

$$\text{IBB}(\mathcal{W}) := \frac{1}{n} \sum_{i=1}^n \sum_{a=1}^C \sum_{y=1}^C \mathbb{I}_{\{y_i=y\}} \phi(F_i, a) \log \left(\frac{\mathbb{I}_{\{y_i=y\}}}{p_y Q(F \in a | Y = y)} \right).$$

The proof of this theorem follows by applying Lemma F.1 and Lemma F.2 in Section F of the supplementary. We remark that $\text{IBB}(\mathcal{W})$ is ready to be optimized by standard SGD algorithms because it is separable and expressed as the summation of losses on individual training points. Algorithm 1 describes the training process of a RIB model for person Re-ID, where $\text{IBB}(\mathcal{W})$ is a regularization term in the training loss. The following functions are needed for minibatch-based training with SGD, with the subscript j indicating the corresponding loss on the j -th batch $\mathcal{B}_j$, and the superscript (t) indicating the corresponding loss at the t -th epoch:

$$\text{IBB}_j^{(t)}(\mathcal{W}) = \frac{1}{|\mathcal{B}_j|} \sum_{i \in \mathcal{B}_j} \sum_{a=1}^C \sum_{y=1}^C \mathbb{I}_{\{y_i=y\}} \phi(F_i, a) \log \left(\frac{\mathbb{I}_{\{y_i=y\}}}{p_y Q^{(t)}(F \in a | Y = y)} \right),$$

where

$$\mathcal{L}_{\text{train},j}^{(t)}(\mathcal{W}) = \text{CE}_j^{(t)} + \text{Triplet}_j^{(t)} + \eta \text{IBB}_j^{(t)}(\mathcal{W}), \quad (2)$$

$$\text{CE}_j^{(t)} = \frac{1}{|\mathcal{B}_j|} \sum_{i \in \mathcal{B}_j} H(F_i(\mathcal{W}), Y_i),$$

$$\text{Triplet}_j^{(t)} = \frac{1}{|\mathcal{B}_j|} \sum_{i \in \mathcal{B}_j} \max(\|F_i - F_i^p\|_2 - \|F_i - F_i^n\|_2, 0).$$

Here, $Q^{(t)}(F \in a | Y = y)$ denotes the variational conditional probability that feature F belongs to cluster a given class label y at epoch t , computed efficiently via Algorithm 3 in the supplementary. $\text{CE}_j^{(t)}$ is the cross-entropy loss on batch $\mathcal{B}_j$ at epoch t , where $H(\cdot, \cdot)$ denotes the cross-entropy function. η is the balance factor for IBB, selected by cross-validation on each dataset. $\text{Triplet}_j^{(t)}$ is the triplet loss [Chen et al., 2017] on batch $\mathcal{B}_j$ at epoch t . $\|\cdot\|_2$ denotes the Euclidean norm. F_i^p and F_i^n denote features from another image of the same person and a different person as F_i , respectively. Let T_0 denote the computational complexity of a forward and backward pass of the neural network with respect to each training sample. The overall complexity for calculating IBB($\mathcal{W}$) is $\Theta(nCT_0 + C^2) = \Theta(nCT_0)$ with $n \geq C$. In contrast, computing the upper bound for the mutual information required for calculating the upper bound for the IB loss proposed in CLUB [Cheng et al., 2020] requires a substantially higher computational complexity of $\Theta(n^2T_0)$. Note that $\Theta(n^2T_0)$ corresponds exclusively to the upper bound for the mutual information $I(F, X)$, while CLUB additionally requires the computation of the lower bound of the mutual information $I(F, Y)$. Details on the complexity analysis of CLUB and IBB are presented in Section D of the supplementary. Moreover, Table 3 in Section 4.4 demonstrates that IBB achieves substantially better performance than CLUB and other competing methods that rely on the unrealistic Gaussian distribution assumption of hidden features [Dai et al., 2018, Srivastava et al., 2021] or on the approximation to the IB loss [Guo et al., 2023].

3.2 APPLICATION OF RIB TO SELF-ATTENTION AND CHANNEL ATTENTION

By adding IBB to the training loss in Equation (2), RIB can be applied to DNNs with self-attention or channel attention modules to promote learning informative attention weights, leading to RIB-DCS and RIB-CA, respectively. To apply RIB to self-attention, we propose learning channel selection masks over the features used to compute the attention weight matrix. As a result, RIB explicitly encourages the use of more informative channels to learn more informative attention weights than vanilla self-attention modules. The channel selection masks are learned in a differentiable manner with a novel Differentiable Channel Selection Attention module, or DCS-Attention, so that they can be jointly optimized with the network parameters, leading to competitive DNNs termed RIB-DCS. RIB can also be applied to DNNs

Algorithm 1 Training Algorithm of the RIB models

Input: Training data $\{X_i, y_i\}_{i=1}^n$, mini-batches of the training data $\{\mathcal{B}_j\}_{j=1}^J$ the epoch number t_{train} , and learning rate α .

Output: The trained weights $\mathcal{W}$ of the network.

- 1: Initialize the weights of the network $\mathcal{W}$ through random initialization.
- 2: Initialize $Q^{(0)}(F|Y) = \frac{1}{C} \mathbf{1}^{C \times C}$.
- 3: Compute $\{\mathcal{C}_b\}_{b=1}^C$ by $\mathcal{C}_b = \frac{\sum_{i=1}^n \mathbf{1}_{\{y_i=b\}} X_i}{\sum_{i=1}^n \mathbf{1}_{\{y_i=b\}}}$.
- 4: **for** $t \leftarrow 1$ to t_{train} **do**
- 5: **for** $j \leftarrow 1$ to J **do**
- 6: Update $\mathcal{W} \leftarrow \mathcal{W} - \alpha \nabla_{\mathcal{W}} \mathcal{L}_{\text{train},j}^{(t)}(\mathcal{W})$ using mini-batch gradient descent on batch $\mathcal{B}_j$.
- 7: **end for**
- 8: Compute $Q^{(t)}(F|Y)$ by Algorithm 3 in Section D of the supplementary.
- 9: Compute $\{\mathcal{C}_a\}_{a=1}^C$ by $\mathcal{C}_a = \frac{\sum_{i=1}^n \mathbf{1}_{\{y_i=a\}} F_i(\mathcal{W})}{\sum_{i=1}^n \mathbf{1}_{\{y_i=a\}}}$.
- 10: **end for**
- 11: **return** The trained weights $\mathcal{W}$ of the network.

with existing channel attention modules by adding IBB to the training loss, leading to RIB-CA.

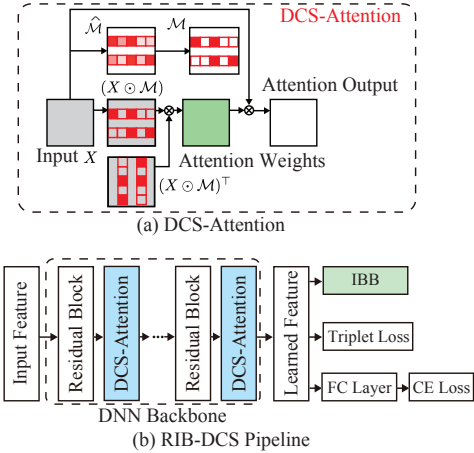


Figure 2: Figure (a) illustrates the framework of the DCS-Attention module. Figure (b) illustrates the pipeline for the RIB-DCS model. The DCS-Attention modules are inserted after each residual block. In a RIB-CA model, the DCS-Attention modules are replaced with existing channel attention modules. IBB is added as a regularization term in addition to the triplet loss and the cross-entropy (CE) loss.

RIB-DCS: RIB with Differentiable Channel Selection for Attention Weight. To explain channel selection for computing attention weights, we first review self-attention modules. Let X be an input feature of shape $N \times D$, where N is the number of feature tokens and D is the number of feature dimensions. For a three-dimensional CNN feature map of

shape $H \times W \times D$, it is reshaped into $N \times D$ with $N = HW$. The vanilla self-attention module applies three linear layers to X to obtain the query (Q), key (K), and value (V), and computes $\text{Output}(Q, K, V) = \text{softmax}(QK^\top)V$. In computer vision tasks, self-attention modules such as non-local self-attention [Wang et al., 2018] are formulated as $\text{Output} = f(g_q(X), g_k(X))g_v(X)$, where $g_q(\cdot)$, $g_k(\cdot)$, and $g_v(\cdot)$ are transformations on X , e.g., convolution or linear layers, and $f(\cdot, \cdot)$ measures feature correlation, such as dot-product or cosine similarity. Recent work [Li et al., 2020] shows that $g_q(\cdot)$, $g_k(\cdot)$, and $g_v(\cdot)$ can be removed due to the strong approximation capability of DNNs. Following [Li et al., 2020], we use $\text{Output} = \frac{1}{C(X)}\sigma(XX^\top)X$, where $C(X)$ is a normalization factor, and $C(X) = 1$ when $\sigma(\cdot)$ is the Softmax operation.

Under the general framework of RIB, we aim to select informative channels to compute the attention weights. To this end, we propose DCS-Attention, which performs differentiable channel selection in the features used for the computation of the attention weights. The channel selection is performed by maintaining a binary decision mask $\mathcal{M} \in \{0, 1\}^{N \times D}$, where its (i, d) -th element $\mathcal{M}_{id} = 1$ indicates the d -th channel of X_i is selected. X_i is the i -th row of X . As a result, the attention weights A are computed by $A = \sigma((X \odot \mathcal{M})(X \odot \mathcal{M})^\top)$, where $\odot$ indicates the elementwise product. To optimize the discrete binary decision mask with gradient descent, we adopt a simplified binary Gumbel-Softmax [Verelst and Tuytelaars, 2020] to relax $\mathcal{M} \in \{0, 1\}^{N \times D}$ into its approximation in the continuous domain $\widehat{\mathcal{M}} \in (0, 1)^{N \times D}$. The approximated soft decision mask can be computed by $\widehat{\mathcal{M}}_{id} = \phi\left(\frac{\theta_{id} + \epsilon_{id}^{(1)} - \epsilon_{id}^{(2)}}{\tau}\right)$, where $\widehat{\mathcal{M}}_{id}$ is the (i, d) -th element of $\widehat{\mathcal{M}}$. $\epsilon_{id}^{(1)}$ and $\epsilon_{id}^{(2)}$ are Gumbel noise terms for the approximation. τ is the temperature, and $\phi(\cdot)$ is the sigmoid function. $\theta \in \mathbb{R}^{N \times D}$ is the sampling parameter. In this work, we obtain θ by applying a linear layer on X so that the soft decision mask is dependent on the input features of that layer. Inspired by the straight-through estimator [Verelst and Tuytelaars, 2020], we directly set $\mathcal{M} = \widehat{\mathcal{M}}$ in the backward pass. In the forward pass, the binary decision mask is computed by $\mathcal{M}_{id} = 1$ if $\widehat{\mathcal{M}}_{id} > 0.5$, and $\mathcal{M}_{id} = 0$ if $\widehat{\mathcal{M}}_{id} \leq 0.5$. During inference, the Gumbel noise $\epsilon_{id}^{(1)}$ and $\epsilon_{id}^{(2)}$ are set to 0. In this way, the informative channels for the attention weights computation are selected in a differentiable manner. To encourage the channel selection masks to select more informative channels in the features for computing the attention weights, the training of the models with DCS-Attention is supervised by the loss function defined in Equation (2), leading to RIB-DCS. Fig. 2(a) illustrates the framework of the DCS-Attention module. Fig. 2(b) illustrates the pipeline of the RIB-DCS model.

RIB-CA: RIB with Existing Channel Attention Methods. Similar to existing self-attention modules that learn

attention weights to quantify the importance of feature tokens, channel attention modules learn attention weights for reweighting different feature channels. However, existing channel attention modules, such as Squeeze-and-Excitation (SE) [Hu et al., 2018], Convolutional Block Attention Module (CBAM) [Woo et al., 2018], and Moment Channel Attention (MCA) [Jiang et al., 2024], learn attention weights using activation statistics or heuristic gating functions. As a result, existing channel attention modules do not explicitly ensure that the channels with higher attention weights are more informative for the discriminative task of person Re-ID. To address this issue, RIB is applied to DNNs equipped with existing channel attention modules, including SE, CBAM, and MCA, in this paper, which leads to RIB-CA. Similar to RIB-DCS, the training of RIB-CA is performed by minimizing the loss function defined in Equation (2).

3.3 RIB WITH FIXED BACKBONE AND LEARNABLE BACKBONE

To evaluate the performance of the proposed RIB, we incorporate RIB-DCS and RIB-CA into both Fixed Backbones (FB), which lead to RIB-DCS-FB and RIB-CA-FB, and learnable backbones with the Differentiable Neural Architecture Search (DNAS) algorithm, which lead to RIB-DCS-DNAS and RIB-CA-DNAS. For RIB-DCS-DNAS models, we first apply the standard DNAS algorithm in FBNetV2 to search for the architectures of the neural network backbones. In the search phase of the RIB-DCS-DNAS models, the weights of the neural network and a set of architecture parameters characterizing the selection of different architecture options in the supernet are jointly optimized. Let $\mathcal{W}$ denote the weights of the neural network, and $\mathcal{V}$ denote the architecture parameters in the supernet. The search loss of RIB-DCS-DNAS and RIB-CA-DNAS models at the t -th epoch on the j -th batch $\mathcal{B}_j$ is

$$\mathcal{L}_{\text{search},j}^{(t)}(\mathcal{W}, \mathcal{V}) = \mathcal{L}_{\text{train},j}^{(t)}(\mathcal{W}, \mathcal{V}) + \mathcal{L}_{\text{latency}}(\mathcal{V}), \quad (3)$$

where $\mathcal{L}_{\text{latency}}(\mathcal{V})$ is the latency cost of the network. We use the same latency cost as in FBNetV2 [Wan et al., 2020]. $\mathcal{L}_{\text{train},j}^{(t)}(\mathcal{W}, \mathcal{V})$ is the training loss in Equation (2). Once the search process is finished, the RIB-DCS-DNAS and RIB-CA-DNAS models with the searched architectures will be retrained. For both the search and the retraining of the RIB-DCS-DNAS and RIB-CA-DNAS models, minibatch-based gradient descent algorithms are used to optimize the network parameters by minimizing the joint training loss $\mathcal{L}_{\text{train},j}^{(t)}(\mathcal{W})$ from Equation (2) for the data in batch $\mathcal{B}_j$ at the t -th epoch.

4 EXPERIMENTS

In this section, we evaluate RIB models on public person Re-ID benchmarks. Section 4.1 introduces the datasets and evaluation metrics. Section 4.2 reports results of RIB-DCS,

Table 1: Performance of RIB-DCS-FB models with comparisons to state-of-the-art Re-ID models when the feature backbones are pre-trained on ImageNet. Smaller-sized backbone enjoys less FLOPs while larger-sized backbones, such as TransReID, exhibit SOTA performance. Results of RIB-DCS-FB models trained from scratch are presented in Table 8 in Section A.5.

Methods	Backbones	Input Size	Params(M)	FLOPs(G)	Market1501		DukeMTMC		MSMT17	
					mAP	Rank-1	mAP	Rank-1	mAP	Rank-1
MobileNetV2 [Sandler et al., 2018]	MobileNetV2	256×128	2.22	0.380	84.5	93.9	72.3	85.5	47.6	74.9
MobileNetV2-200 [Sandler et al., 2018]	MobileNetV2-200	256×128	5.04	0.890	85.0	94.5	75.2	87.4	50.0	76.3
ResNet50 [He et al., 2016]	ResNet50	256×128	23.5	6.02	86.8	95.5	78.1	88.5	57.3	81.6
AANet [Tay et al., 2019]	ResNet50	256×128	27.1	7.36	85.3	94.7	75.3	84.0	58.6	81.5
CAMA [Yang et al., 2019]	ResNet50	256×128	28.4	7.85	84.5	94.7	72.9	85.8	57.4	79.5
BAT-Net [Fang et al., 2019]	ResNet50	256×128	26.8	7.54	87.4	95.1	77.3	87.7	58.8	81.0
ABD-Net [Chen et al., 2019]	ResNet50	384×128	69.2	14.1	88.3	95.6	78.6	89.0	60.8	82.3
Auto-ReID [Quan et al., 2019]	ResNet50	384×128	13.1	2.05	85.1	94.5	77.6	88.8	52.5	78.2
OSNet [Zhou et al., 2019]	OSNet	256×128	2.2	0.98	84.9	94.8	73.5	88.6	52.9	78.7
RGA [Zhang et al., 2020]	ResNet50	256×128	28.3	7.10	87.5	96.0	77.4	88.8	57.5	80.3
PAT [Li et al., 2021b]	ResNet50	256×128	27.8	7.95	88.0	95.4	78.2	88.8	58.9	81.0
AutoLoss-GMS [Gu et al., 2022]	ResNet50	256×128	23.9	6.89	87.0	94.7	78.5	89.2	55.1	79.5
UAL [Dou et al., 2022]	ResNet50	256×128	24.0	7.5	87.0	95.2	80.0	89.5	56.5	80.0
UAL [Dou et al., 2022]	HRNet	256×128	30.5	9.0	89.5	95.7	82.5	90.8	65.3	84.7
TransReID [He et al., 2021]	ViT-B/16	256×128	100	19.3	88.9	95.2	82.0	90.7	67.4	85.3
BPBreID [Somers et al., 2023]	ResNet50	256×128	24.4	6.9	87.0	95.1	78.3	89.6	57.0	80.9
PFID [Wang et al., 2022]	TransReID	256×128	100	19.3	88.7	95.5	83.2	91.2	68.5	85.8
PHA [Zhang et al., 2023]	TransReID	256×128	100	19.3	<u>90.3</u>	96.1	83.0	91.0	<u>68.9</u>	<u>86.1</u>
RIB-DCS-FB (MobileNetV2)	MobileNetV2	256×128	2.31	0.406	86.4 (↑ 1.9)	95.1 (↑ 1.2)	75.6 (↑ 3.3)	86.9 (↑ 1.4)	53.1 (↑ 5.5)	78.5 (↑ 3.6)
RIB-DCS-FB (MobileNetV2-200)	MobileNetV2-200	256×128	5.32	0.927	87.5 (↑ 2.5)	95.6 (↑ 1.1)	78.6 (↑ 3.4)	89.1 (↑ 1.7)	54.8 (↑ 4.8)	78.9 (↑ 2.6)
RIB-DCS-FB (ResNet50)	ResNet50	256×128	23.9	6.24	88.7 (↑ 1.9)	96.0 (↑ 0.5)	78.8 (↑ 0.7)	88.9 (↑ 0.4)	58.9 (↑ 1.6)	82.7 (↑ 1.1)
RIB-DCS-FB (OSNet)	OSNet	256×128	2.33	1.03	86.8 (↑ 1.9)	94.9 (↑ 0.1)	75.8 (↑ 2.3)	89.0 (↑ 0.4)	55.1 (↑ 2.2)	79.8 (↑ 1.1)
RIB-DCS-FB (HRNet)	HRNet	256×128	32.1	9.9	90.1 (↑ 0.6)	96.2 (↑ 0.5)	82.8 (↑ 0.5)	91.1 (↑ 0.3)	66.9 (↑ 1.6)	85.1 (↑ 0.4)
RIB-DCS-FB (TransReID)	TransReID	256×128	101.6	19.7	91.3 (↑ 2.4)	97.0 (↑ 1.8)	84.3 (↑ 2.3)	92.0 (↑ 1.3)	70.2 (↑ 2.8)	87.1 (↑ 1.8)

and Section 4.3 presents results of RIB-CA. Section 4.4 compares the efficiency and performance of the proposed IBB with existing IB loss reduction methods. Section 4.5 evaluates RIB on occluded person Re-ID. Additional results are provided in Section A of the supplementary. Dataset details and training settings of DCS models are given in Sections A.1 and A.2. The supernet structures of FBNetV2 variants are described in Section A.3. Section A.4 presents ablation studies on IBB. Results of RIB-DCS-FB trained from scratch are shown in Section A.5. Results of RIB-DCS-DNAS and RIB-CA-DNAS are reported in Sections A.6 and A.7. Training time and search time analyses are provided in Sections A.8 and A.9. Section A.10 compares RIB-DCS with an existing attention masking method, MODA [Zhang et al., 2025]. Section A.11 reports results on CUHK03. Section A.12 evaluates cross-domain performance with CaCL [Lee et al., 2023]. Section A.13 studies RIB in self-supervised Re-ID. Section A.14 analyzes the DCS-Attention modules. Section A.15 provides additional GradCAM visualizations. t-SNE analysis is presented in Section A.16 of the supplementary. Loss curves of IBB and test loss are shown in Section C of the supplementary. Section E demonstrates the advantages of hard channel selection in DCS-Attention. Section A.17 compares RIB with recent state-of-the-art Re-ID methods, PersonViT [Hu et al., 2025] and HPL [Zhou et al., 2026]. Section A.18 extends the class-centroid modeling of RIB to prototypical learning, leading to RIBPL, and studies its sensitivity to the number of prototypes. Section A.19 presents a quantitative evaluation of the attention quality and the controlled effect of the balance factor η for IBB. Section A.20 ablates the contributions of the loss terms,

and Section A.21 ablates the individual contributions of IBB and DCS-Attention. Section A.22 reports the inference time of RIB models. Section G analyzes the consistency of the mutual information estimators used in RIB, and Section H discusses the limitations of RIB.

4.1 DATASETS AND EXPERIMENT SETTINGS

We evaluate the RIB models on four public person Re-ID datasets, which are Market-1501 [Zheng et al., 2015], DukeMTMC [Ristani et al., 2016], MSMT17 [Wei et al., 2018], and CUHK03 [Li et al., 2014]. Standard Re-ID metrics Rank-1 accuracy and the mean Average Precision (mAP) are used. Re-ranking [Zhong et al., 2017] and multi-query fusion [Zheng et al., 2015] are not used for the fairness of comparison. The statistics of the evaluation datasets, the training settings of the DCS models, and the tuning details of the balance factor η for IBB together with the supernet structures of FBNetV2, FBNetV2-Large, and FBNetV2-XLarge are provided in Sections A.1, A.2, and A.3 of the supplementary, respectively.

4.2 RIB-DCS FOR PERSON RE-ID

We compare the performance of our RIB-DCS-FB models with existing state-of-the-art (SOTA) methods for person Re-ID on three datasets. Table 1 shows the performance of RIB-DCS-FB models based on MobileNetV2 [Sandler et al., 2018], OSNet [Zhou et al., 2019], ResNet50 [He et al., 2016], HRNet [Dou et al., 2022], and TransReID [Wang

Table 2: Performance of RIB-CA-FB models with the state-of-the-art fixed backbone, TransReID, on Market1501, DukeMTMC, and MSMT17.

Method	Params(M)	FLOPs(G)	Market1501		DukeMTMC		MSMT17	
			mAP	Rank-1	mAP	Rank-1	mAP	Rank-1
TransReID+SE [Hu et al., 2018]	100.6	19.5	89.5	95.6	82.6	90.9	68.0	85.5
RIB-CA-FB (TransReID+SE)	100.8	19.8	90.6 (↑ 1.1)	96.6 (↑ 1.0)	83.7 (↑ 1.1)	91.8 (↑ 0.9)	69.1 (↑ 1.1)	86.5 (↑ 1.0)
TransReID+CBAM [Woo et al., 2018]	100.9	19.7	89.7	95.8	82.8	91.1	68.4	85.9
RIB-CA-FB (TransReID+CBAM)	101.3	20.0	90.9 (↑ 1.2)	96.7 (↑ 0.9)	83.9 (↑ 1.1)	92.0 (↑ 0.9)	69.5 (↑ 1.1)	87.0 (↑ 1.1)
TransReID+MCA [Jiang et al., 2024]	101.6	19.9	90.1	96.0	83.0	91.2	68.4	85.7
RIB-CA-FB (TransReID+MCA)	102.1	20.3	91.0 (↑ 0.9)	96.8 (↑ 0.8)	84.1 (↑ 1.1)	91.9 (↑ 0.7)	69.5 (↑ 1.1)	86.8 (↑ 1.1)

et al., 2022] on all three person Re-ID datasets. The RIB-DCS-FB models based on the above backbones are referred to as RIB-DCS-FB where the corresponding backbone name is in the parentheses. We evaluate the performance of all RIB-DCS-FB models with backbones pretrained on ImageNet. For the compact RIB-DCS-FB models, including RIB-DCS-FB (OSNet), RIB-DCS-FB (MobileNetV2), and RIB-DCS-FB (MobileNetv2-200), we also evaluate their performance by training them from scratch on the person Re-ID datasets. It is observed in Table 1 that all the RIB-DCS-FB models outperform existing SOTA methods using the same corresponding feature learning backbones. The best RIB-DCS-FB model, RIB-DCS-FB (TransReID), achieves SOTA results on all three person Re-ID datasets. For example, RIB-DCS-FB (TransReID) outperforms PHA, which also uses TransReID as the feature learning backbone, by 1.3% in mAP for person Re-ID on MSMT17. The results for RIB-DCS-DNAS are deferred to Table 9 in Section A.6 of the supplementary.

4.3 RIB-CA FOR PERSON RE-ID

We apply RIB-CA to the SOTA person Re-ID backbone, TransReID [He et al., 2021]. We first insert existing channel attention modules, including Squeeze-and-Excitation (SE) [Hu et al., 2018], Convolutional Block Attention Module (CBAM) [Woo et al., 2018], and Moment Channel Attention (MCA) [Jiang et al., 2024], after each transformer block in TransReID, leading to TransReID+SE, TransReID+CBAM, and TransReID+MCA. The RIB-CA models based on TransReID+SE, TransReID+CBAM, and TransReID+MCA are denoted as RIB-CA-FB (TransReID+SE), RIB-CA-FB (TransReID+CBAM), and RIB-CA-FB (TransReID+MCA). It is observed in Table 2 that the RIB-CA-FB models consistently outperform their corresponding baseline models across all the person Re-ID benchmarks. For instance, RIB-CA-FB (TransReID+CBAM) outperforms TransReID+CBAM by 1.2% in mAP for person Re-ID on Market1501, demonstrating the effectiveness of learning informative channel attention weights with RIB-CA. The results for RIB-CA-DNAS are deferred to Table 10 in Section A.7 of the supplementary.

4.4 COMPARISON BETWEEN IBB AND EXISTING UPPER BOUNDS FOR THE IB LOSS

We compare the proposed variational upper bound for the IB loss, IBB, with existing works deriving the upper bound for the IB loss [Cheng et al., 2020, Dai et al., 2018, Srivastava et al., 2021]. Although CLUB [Cheng et al., 2020] also proposes an upper bound for the IB loss, CLUB assumes that $p(F|X)$ is known and adopts a Gaussian Mixture Model (GMM) parameterized by an external neural network to approximate the upper bound when $p(F|X)$ is unknown. As shown in Lemma F.1 in Section F of the supplementary, $p(F|X)$ in our formulation can be directly computed from the training data without the need for training another neural network. The upper bound for the IB loss proposed in VIB [Dai et al., 2018, Srivastava et al., 2021] imposes an unrealistic Gaussian distribution assumption on the hidden features of DNNs. Instead of reducing the IB loss or its upper bound, APIB [Guo et al., 2023] reduces only an approximation to the IB loss. To compare the performance of IBB with VIB, APIB, and CLUB, we conduct an ablation study by replacing IBB with VIB, APIB, and CLUB in RIB-DCS-FB (TransReID) for person Re-ID on Market1501, with the total training time on a single NVIDIA A100 GPU reported. It is observed in Table 3 that the unrealistic Gaussian distribution assumption on the hidden features imposed by VIB [Dai et al., 2018] and the IB approximation in APIB [Guo et al., 2023] lead to degraded performance compared with our IBB. For instance, the model based on IBB outperforms the model based on VIB by 1.4% in mAP on Market1501. In addition, the model based on IBB outperforms the model based on CLUB by 1.0% in mAP while using only 56.1% of the training time of the CLUB-based model.

Table 3: Comparison of different methods for reducing the IB loss. The study is performed on Market1501 with the feature backbone RIB-DCS-FB (TransReID).

Methods	Training Time (Hours)	mAP	Rank-1
VIB [Dai et al., 2018]	32.2	89.9	95.8
APIB [Guo et al., 2023]	32.6	90.1	96.0
CLUB [Cheng et al., 2020]	58.6	90.3	95.9
IBB (Ours)	32.9	91.3	97.0

4.5 EVALUATION ON OCCLUDED-DUKE

To further verify the robustness of the proposed method for person Re-ID under occlusion, we conduct experiments on the Occluded-Duke [Miao et al., 2019] dataset. Images in Occluded-Duke contain severe partial occlusions and spatial misalignments caused by diverse viewing angles and pedestrian overlaps, making person re-identification particularly challenging. We evaluate both TransReID [He et al., 2021] and SPT [Tan et al., 2024], along with the RIB-DCS-FB models using TransReID and SPT as backbones, under the same training and inference configuration described in Section A.2. SPT is the state-of-the-art occluded person Re-ID model based on a transformer architecture that uses vanilla self-attention modules similar to TransReID. It is observed in Table 4 that RIB-DCS significantly improves the performance of both TransReID and SPT, consistently achieving higher mAP and Rank-1 accuracy. For example, RIB-DCS-FB (SPT) outperforms SPT by 1.2% in mAP, which demonstrates that RIB effectively learns informative and semantically relevant representations that are robust to the occlusion and background in the input image.

Table 4: Performance comparison on Occluded-Duke under the same training and inference settings without re-ranking.

Method	mAP	Rank-1
TransReID [He et al., 2021]	59.2	66.4
RIB-DCS-FB (TransReID)	60.5	67.9
SPT [Tan et al., 2024]	63.0	74.7
RIB-DCS-FB (SPT)	64.2	75.7

5 CONCLUSION

In this paper, we propose a novel distribution-free and efficient IB reduction framework, termed RIB. Motivated by the IB principle, RIB learns informative attention weights for DNNs with self-attention and channel attention modules by explicitly reducing the IB loss. We derive and prove a distribution-free and efficient variational upper bound for the IB loss, termed IBB, which can be optimized by standard SGD. To incorporate RIB into self-attention modules, we develop a Differentiable Channel Selection Attention method, or DCS-Attention, which selects more informative feature channels to compute more informative attention weights than vanilla self-attention, resulting in RIB-DCS. RIB is also applied to DNNs with existing channel attention modules, yielding RIB-CA that promotes informative channel attention weights. Both RIB-DCS and RIB-CA are implemented on fixed and learnable backbones. Extensive experiments, including occluded person Re-ID, cross-domain person Re-ID, and self-supervised person Re-ID, demonstrate that RIB consistently improves the performance of DNNs with self-attention and channel attention.

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Learning Informative Attention Weights for Person Re-Identification (Supplementary Material)

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A ADDITIONAL EXPERIMENT RESULTS

A.1 DETAILS ON THE DATASETS

Market-1501 [Zheng et al., 2015] consists of 32,668 annotated person images of 1,501 identities, in which 12,936 images of 751 identities are used for training and 19,732 images of 750 identities are used for testing. All images are shot with 6 different cameras. DukeMTMC [Ristani et al., 2016] was originally proposed for video-based Re-ID. It has 16,522 training images of 702 identities. The other 2,228 query images of 702 identities and 17,661 gallery images are in the test set. MSMT17 [Wei et al., 2018] is the largest and the most challenging public person re-ID dataset, which includes 126,441 person images of 4,101 identities detected by Faster R-CNN [Ren et al., 2015]. The training set has 32,621 person images of 1,041 identities, and the test set has 93,820 images of the other 3,060 identities. CUHK03 [Li et al., 2014] consists of 13,164 person images of 1,467 identities, of which images of 767 identities are in the training set, and the remaining 700 identities are in the test set.

A.2 DETAILED TRAINING SETTINGS

Training Settings of RIB Models with Fixed Backbones, including RIB-DCS-FB and RIB-CA-FB. Following the settings in existing works [He et al., 2021, Zhang et al., 2020] for person Re-ID, random cropping, horizontal flipping, and random erasing are used to augment the training data. Label smoothing [Szegedy et al., 2016] is used to augment the labels of the cross-entropy loss. All models are trained with momentum SGD for 600 epochs. The batch size is set to 256. The momentum for SGD is set to 0.9. The initial learning rate is set to 0.035, and we decay the learning rate by 10 at the 300-th and 500-th epochs. We set the weight decay of SGD to 0.0005. To stabilize the training of DCS-Attention modules, we include a warm-up process that disables channel selection for the first 10 epochs.

Training Settings of RIB Models with Learnable Backbones. The search phase of RIB-DCS-DNAS and RIB-CA-DNAS models take 300 epochs. In each epoch, the network weights are trained with 80% of the training samples by SGD. The architecture parameters are trained with the remaining 20% using Adam. The initial learning rate for optimizing architecture parameters is set to 0.03, and a cosine learning rate schedule is used. The initial learning rate for network parameters is set to 0.035, and it is decayed by 10 at the 150-th and 240-th epochs. After the search phase, we first pre-train the model with the searched architectures on ImageNet. Then we fine-tune the model on the Re-ID datasets. Adam is used to fine-tune the network. The initial learning rate is set to 0.00035. All models are fine-tuned for 600 epochs. The learning rate is decayed by 10 after 300 and 500 epochs. In our work, we build RIB-DCS-DNAS models based on FBNetV2 [Wan et al., 2020].

A.3 ADDITIONAL IMPLEMENTATION DETAILS

Tuning Hyper-Parameters by Cross-Validation. To decide the best balancing factor η for the IBB in the overall training loss, we perform 5-fold cross-validation on 10% of the training data. The value of η is selected from $\{0.1, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 2\}$. The validation results suggest that $\eta = 1$ works best for all our RIB-DCS

and RIB-CA models on different datasets.

FBNetV2 Supernet Structures. FBNetV2 [Wan et al., 2020] employs the Differentiable Neural Architecture Search (DNAS) algorithm to learn the backbone architecture by selecting neural options in a supernet. The search space of FBNetV2 features a masking mechanism based on Gumbel-Softmax for feature map reuse so that it can efficiently search for the number of filters of each convolution layer. In addition, we designed two larger supernet architectures of FBNetV2, termed FBNetV2-Large and FBNetV2-XLarge. The depth of FBNetV2-Large is 3 times the depth of the original supernet of FBNetV2. The depth of FBNetV2-XLarge is 6 times the depth of the original supernet of FBNetV2. The structure of the supernet of FBNetV2, FBNetV2-Large, and FBNetV2-XLarge are detailed in Table 5 and Table 6, respectively.

Table 5: The supergraph of FBNetV2 and FBNetV2-Large ($3 \times$ depth), with block expansion rate e , number of filters f , number of blocks n , and stride of first block s . Tuples of three values in the column of expansion rate e and number of filters f represent the lowest value, highest, and steps between options (low, high, steps).

Operator	e	f	n		s
			FBNetV2	FBNetV2-Large	
conv2d	1	16	1	1	2
bottleneck	1	(12, 16, 4)	1	1	1
bottleneck	(0.75, 3.25, 0.5)	(16, 28, 4)	1	1	2
bottleneck	(0.75, 3.25, 0.5)	(16, 28, 4)	2	6	1
DCS-Attention Module or Channel Attention Module Inserted					
bottleneck	(0.75, 3.25, 0.5)	(16, 40, 8)	1	3	2
bottleneck	(0.75, 3.25, 0.5)	(16, 40, 8)	2	6	1
DCS-Attention Module or Channel Attention Module Inserted					
bottleneck	(0.75, 3.25, 0.5)	(48, 96, 8)	1	3	2
bottleneck	(0.75, 3.25, 0.5)	(48, 96, 8)	2	6	1
bottleneck	(0.75, 4.5, 0.75)	(72, 128, 8)	4	12	1
DCS-Attention Module or Channel Attention Module Inserted					
bottleneck	(0.75, 4.5, 0.75)	(112, 216, 8)	1	3	2
bottleneck	(0.75, 4.5, 0.75)	(112, 216, 8)	3	3	1
DCS-Attention Module or Channel Attention Module Inserted					
conv2d	-	1984	1	1	1
fc	-	-	1	-	-

Table 6: The supergraph of FBNetV2-XLarge ($6 \times$ depth), with block expansion rate e , number of filters f , number of blocks n , and stride of first block s . Tuples of three values in the column of expansion rate e and number of filters f represent the lowest value, highest, and steps between options (low, high, steps).

Operator	e	f	n	s
conv2d	1	16	1	2
bottleneck	1	(12, 16, 4)	1	1
bottleneck	(0.75, 6.25, 0.5)	(16, 28, 4)	1	2
bottleneck	(0.75, 6.25, 0.5)	(16, 28, 4)	12	1
DCS-Attention Module or Channel Attention Module Inserted				
bottleneck	(0.75, 6.25, 0.5)	(16, 40, 8)	6	2
bottleneck	(0.75, 6.25, 0.5)	(16, 40, 8)	12	1
DCS-Attention Module or Channel Attention Module Inserted				
bottleneck	(0.75, 6.25, 0.5)	(48, 96, 8)	6	2
bottleneck	(0.75, 6.25, 0.5)	(48, 96, 8)	12	1
bottleneck	(0.75, 7.5, 0.75)	(72, 128, 8)	24	1
DCS-Attention Module or Channel Attention Module Inserted				
bottleneck	(0.75, 7.5, 0.75)	(112, 216, 8)	6	2
bottleneck	(0.75, 7.5, 0.75)	(112, 216, 8)	6	1
DCS-Attention Module or Channel Attention Module Inserted				
conv2d	-	1984	1	1
avgpool	-	-	1	1
fc	-	-	-	-

A.4 ABLATION STUDY ON THE EFFECTS OF IBB

In this section, we study the effectiveness of the variational upper bound for the IB loss, which is IBB, in the training of RIB-DCS-FB and RIB-DCS-DNAS models for person Re-ID. The comparisons are performed on Market1501 with three base models, which are MobileNetV2, TransReID, and FBNetV2-Large. Since the vanilla MobileNetV2 and FBNetV2-Large do not contain attention modules, we add two baseline models that incorporate vanilla self-attention modules into MobileNetV2 and FBNetV2-Large, which are referred to as MobileNetV2+Self-Attention and FBNetV2-Large+Self-Attention. We also evaluate three ablation models, which incorporate the DCS-Attention modules into the backbones of MobileNetV2, TransReID, and FBNetV2-Large. The above ablation models that only incorporate DCS-Attention without reducing IBB are referred to as MobileNetV2+DCS-Attention, TransReID+DCS-Attention, and FBNetV2-Large+DCS-Attention, respectively. It is observed in Table 7 that although optimizing the regular cross-entropy loss and the triplet loss to train DNNs with DCS-Attention can decrease the IB loss so that the IB principle, learning features more strongly correlated with class labels while decreasing their correlation with the input, is adhered to, the IB loss can be decreased further by a considerable extent by optimizing the IBB explicitly. In particular, our RIB-DCS-FB and RIB-DCS-DNAS models outperform the baseline models and the ablation models by large margins in terms of IB loss, mAP, and Rank-1 accuracy. For instance, RIB-DCS-FB (TransReID) further decreases the IB loss of TransReID+DCS-Attention by 0.039, leading to a 0.9% increase in mAP and a 1.2% increase in Rank-1 accuracy for person Re-ID on Market1501.

Table 7: Ablation study on the effects of IBB. The comparisons are performed on Market1501.

Model	mAP	Rank-1	IBB	IB Loss
MobileNetV2	84.5	93.9	0.062	-0.018
MobileNetV2+Self-Attention	85.0	94.1	0.060	-0.019
MobileNetV2+DCS-Attention	85.8	94.5	0.055	-0.027
RIB-DCS-FB (MobileNetV2)	86.4	95.1	0.028	-0.064
TransReID	88.9	95.2	0.059	-0.020
TransReID+DCS-Attention	90.4	95.8	0.050	-0.028
RIB-DCS-FB (TransReID)	91.3	97.0	0.027	-0.067
FBNetV2-Large	87.2	95.0	0.061	-0.019
FBNetV2-Large+Self-Attention	87.9	95.2	0.059	-0.020
FBNetV2-Large+DCS-Attention	88.7	95.4	0.053	-0.028
RIB-DCS-DNAS (FBNetV2-Large)	89.1	95.8	0.028	-0.065

A.5 ADDITIONAL RESULTS FOR RIB-DCS-FB TRAINED FROM SCRATCH

In this section, we evaluate RIB-DCS-FB when the backbone networks are trained entirely from scratch and compare them with representative state-of-the-art Re-ID models. It is observed in Table 8 that RIB-DCS-FB consistently improves both mAP and Rank-1 accuracy over their corresponding baselines across all datasets, while introducing only marginal increases in parameters and FLOPs. For instance, on Market1501, RIB-DCS-FB (OSNet) achieves an improvement of 2.7% in mAP and 0.6% in Rank-1 over OSNet, while on DukeMTMC it improves mAP by 5.4%. The gains are even more pronounced on MSMT17, where RIB-DCS-FB (MobileNetV2) improves mAP and Rank-1 by 10.7% and 11.7%, demonstrating that the proposed RIB mechanism enhances feature discriminability and generalization capability even without ImageNet pretraining.

A.6 RESULTS FOR RIB-DCS-DNAS

Differentiable Neural Architecture Search (DNAS) methods have demonstrated considerable potential in deriving neural networks that are both efficient and effective [Wan et al., 2020, Quan et al., 2019]. Therefore, we apply RIB-DCS-DNAS to search for architectures of efficient DNNs incorporating DCS-Attention that achieve compelling performance for person Re-ID. The evaluation is performed with three versions of the supernet based on FBNetV2. For comparison, we also evaluate the performance of the architectures searched with these three versions of supernet without RIB-DCS. We emphasize that RIB is effective without NAS: as shown in Tables 1 and 2 in Sections 4.2 and 4.3, and in the results with fixed backbones in the supplementary, RIB achieves state-of-the-art results without NAS. Incorporating DNAS in this paper is not intended to improve the performance of RIB models. Instead, it is used to study how RIB can be combined with DNAS to learn compact

Table 8: Performance of RIB-DCS-FB models with comparisons to state-of-the-art Re-ID models when the feature backbones are trained from scratch. Smaller-sized backbone enjoys less FLOPs while larger-sized backbones, such as TransReID, exhibit SOTA performance. The improvements of the RIB models over their corresponding baselines are reported in parentheses after each metric value.

Methods	Backbones	Input Size	Params(M)	FLOPs(G)	Market1501		DukeMTMC		MSMT17	
					mAP	Rank-1	mAP	Rank-1	mAP	Rank-1
HACNN [Li et al., 2018]	Inception	160×64	4.5	0.55	79.9	92.3	63.8	80.5	41.5	70.6
OSNet [Zhou et al., 2019]	OSNet	256×128	2.2	0.98	81.0	93.6	68.6	84.7	43.3	71.0
Auto-ReID [Chen et al., 2019]	ResNet50	384×128	13.1	2.05	74.6	90.7	60.8	78.6	32.5	60.5
RGA [Zhang et al., 2020]	MobileNetV2	256×128	5.13	2.63	81.5	92.9	70.5	83.7	36.8	62.1
MobileNetV2 [Sandler et al., 2018]	MobileNetV2	256×128	2.22	0.380	78.9	92.0	68.5	82.7	34.6	61.2
MobileNetV2-200 [Sandler et al., 2018]	MobileNetV2-200	256×128	5.04	0.890	81.8	93.4	71.8	84.5	43.2	68.8
RIB-DCS-FB (OSNet)	OSNet	256×128	2.33	1.03	83.7 (↑ 2.7)	94.2 (↑ 0.6)	74.0 (↑ 5.4)	85.8 (↑ 1.1)	46.5 (↑ 3.2)	73.2 (↑ 2.2)
RIB-DCS-FB (MobileNetV2)	MobileNetV2	256×128	2.31	0.406	82.5 (↑ 3.6)	93.0 (↑ 1.0)	74.5 (↑ 6.0)	86.0 (↑ 3.3)	45.3 (↑ 10.7)	72.9 (↑ 11.7)
RIB-DCS-FB (MobileNetV2-200)	MobileNetV2-200	256×128	5.32	0.927	84.5 (↑ 2.7)	94.5 (↑ 1.1)	74.9 (↑ 3.1)	87.0 (↑ 2.5)	47.0 (↑ 3.8)	73.5 (↑ 4.7)

RIB models, since DNAS has been widely used in the deep learning literature [Wan et al., 2020, Salmani Pour Avval et al., 2025] to search for efficient architectures. For example, RIB-DCS-DNAS (FBNetV2-XLarge), with only 1.898G FLOPs, outperforms TransReID [He et al., 2021] with 19.3G FLOPs by 1.2% in Rank-1 on Market1501, demonstrating the efficiency and effectiveness of RIB-DCS-DNAS. It is observed in Table 9 that all RIB-DCS-DNAS models significantly outperform corresponding baseline DNAS models. For instance, RIB-DCS-DNAS (FBNetV2-XLarge) outperforms FBNetV2-XLarge by 1.6% in mAP on MSMT17. In addition, with a small number of FLOPs, RIB-DCS-DNAS models outperform existing methods that require significantly more computational resources. Notably, RIB-DCS-DNAS (FBNetV2-XLarge), with only 1.898G FLOPs, outperforms ABD-Net with 14.1G FLOPs by 2.1% in mAP on the MSMT17 dataset, demonstrating the efficiency and effectiveness of RIB-DCS-DNAS models.

Table 9: Performance of RIB-DCS-DNAS models on Market1501, DukeMTMC, and MSMT17. The improvements of the RIB models over their corresponding baselines are reported in parentheses after each metric value.

Method	Params(M)	FLOPs(G)	Market1501		DukeMTMC		MSMT17	
			mAP	Rank-1	mAP	Rank-1	mAP	Rank-1
Auto-ReID [Quan et al., 2019]	13.1	2.05	85.1	94.5	77.6	88.8	52.5	78.2
RGA [Zhang et al., 2020]	28.3	7.3	87.5	96.0	77.4	88.8	57.5	80.3
UAL [Dou et al., 2022]	24.0	7.5	87.0	95.2	80.0	89.5	56.5	80.0
BPBreID [Somers et al., 2023]	24.4	6.9	87.0	95.1	78.3	89.6	57.0	80.9
ABD-Net [Chen et al., 2019]	69.2	14.1	88.3	95.6	78.6	89.0	60.8	82.3
FBNetV2	5.9	0.572	84.6	94.1	75.9	87.4	58.1	80.5
RIB-DCS-DNAS (FBNetV2)	6.3	0.624	85.8 (↑ 1.2)	95.3 (↑ 1.2)	77.1 (↑ 1.2)	88.6 (↑ 1.2)	59.5 (↑ 1.4)	81.9 (↑ 1.4)
FBNetV2-Large	13.2	1.235	87.2	95.0	77.8	88.9	59.9	82.5
RIB-DCS-DNAS (FBNetV2-Large)	13.7	1.302	89.1 (↑ 1.9)	95.8 (↑ 0.8)	79.2 (↑ 1.4)	90.2 (↑ 1.3)	61.0 (↑ 1.1)	83.0 (↑ 0.5)
FBNetV2-XLarge	24.5	1.798	87.9	95.3	79.5	90.1	61.3	82.9
RIB-DCS-DNAS (FBNetV2-XLarge)	25.2	1.898	89.8 (↑ 1.9)	96.4 (↑ 1.1)	81.0 (↑ 1.5)	91.4 (↑ 1.3)	62.9 (↑ 1.6)	83.6 (↑ 0.7)

A.7 RESULTS FOR RIB-CA-DNAS

We further evaluate RIB-CA under the setting where existing channel attention modules are integrated into the FBNetV2 supernet, after which the backbone architecture is optimized using the DNAS algorithm. The experiments are performed using the best channel attention module, MCA [Jiang et al., 2024]. The models incorporating MCA based on FBNetV2, FBNetV2-Large, and FBNetV2-XLarge are denoted as FBNetV2+MCA, FBNetV2-Large+MCA, and FBNetV2-XLarge+MCA. It is observed in Table 10 that the RIB-CA-DNAS models consistently outperform the baseline models incorporating existing attention modules into the DNAS backbones. For instance, RIB-CA-DNAS (FBNetV2-XLarge+MCA) outperforms FBNetV2-XLarge+MCA by 1.2% in mAP on Market1501, demonstrating the advantages of learning informative channel attention weights with learnable backbones.

Table 10: Performance of RIB-CA-DNAS with FBNetV2, FBNetV2-Large, and FBNetV2-XLarge on Market1501, DukeMTMC, and MSMT17. The improvements of the RIB models over their corresponding baselines are reported in parentheses after each metric value.

Method	Params(M)	FLOPs(G)	Market1501		DukeMTMC		MSMT17	
			mAP	Rank-1	mAP	Rank-1	mAP	Rank-1
FBNetV2+MCA [Jiang et al., 2024]	5.97	0.586	84.8	94.3	76.2	87.7	58.4	80.8
RIB-CA-DNAS (FBNetV2+MCA)	6.41	0.603	85.6 (↑0.8)	95.1 (↑0.8)	76.9 (↑0.7)	88.8 (↑1.1)	59.3 (↑0.9)	81.8 (↑1.0)
FBNetV2-Large+MCA [Jiang et al., 2024]	13.4	1.262	85.9	95.2	78.0	89.1	60.2	82.5
RIB-CA-DNAS (FBNetV2-Large+MCA)	13.8	1.298	86.8 (↑0.9)	95.9 (↑0.7)	79.0 (↑1.0)	89.9 (↑0.8)	60.9 (↑0.7)	82.9 (↑0.4)
FBNetV2-XLarge+MCA [Jiang et al., 2024]	24.9	1.868	88.5	95.5	79.8	90.2	61.7	82.9
RIB-CA-DNAS (FBNetV2-XLarge+MCA)	25.5	1.985	89.7 (↑1.2)	96.2 (↑0.7)	80.6 (↑0.8)	91.1 (↑0.9)	62.8 (↑1.1)	83.5 (↑0.6)

A.8 TRAINING TIME EVALUATION

In this section, we assess the training costs of our RIB-DCS-FB and RIB-DCS-DNAS models and the corresponding baseline models on Market1501. This evaluation is performed on one NVIDIA A100 GPU, utilizing an effective batch size of 256 images. The total training duration for 600 epochs is reported. It is observed in Table 11 that our RIB-DCS-FB and RIB-DCS-DNAS models only marginally increase the training time of the corresponding baselines while significantly improving the performance. For example, the training time of RIB-DCS-FB (TransReID) is only 5.7% longer than the training time of TransReID, while RIB-DCS-FB (TransReID) outperforms TransReID by 2.4% in mAP on Market1501.

Table 11: Training time (hours) comparisons on Market1501 between RIB-DCS-FB (MobileNetV2), RIB-DCS-FB (TransReID), RIB-DCS-DNAS (FBNetV2-Large), and their corresponding baseline models.

Model	Training Time	mAP	R1
MobileNetV2	10.3	84.5	93.9
RIB-DCS-FB (MobileNetV2)	10.9	86.4	95.1
TransReID	31.1	88.9	95.2
RIB-DCS-FB (TransReID)	32.9	91.3	97.0
FBNetV2-Large	16.7	87.2	95.0
RIB-DCS-DNAS (FBNetV2-Large)	17.5	89.1	95.8

A.9 SEARCH TIME EVALUATION

In this section, we assess the search costs of our RIB-DCS-DNAS and RIB-CA-DNAS models and the corresponding baseline models, which do not incorporate IBB into the training loss. The evaluation is conducted on Market1501 using one NVIDIA A100 GPU. It is observed in Table 12 that our RIB only marginally increases the search time of the baseline models, FBNetV2-XLarge+MCA and FBNetV2-XLarge, while significantly increasing their performance for person ReID. For example, RIB-DCS-DNAS (FBNetV2-XLarge) requires only an additional 7.2% of search time compared to FBNetV2-XLarge, while achieving a significant improvement of 1.2% in mAP.

Table 12: Searching time (hours) comparisons on Market1501 between RIB-CA-DNAS (FBNetV2-XLarge+MCA), RIB-DCS-DNAS (FBNetV2-XLarge) and their corresponding baseline models.

Model	Training Time	mAP	R1
FBNetV2-XLarge+MCA	2.26	87.9	95.3
RIB-CA-DNAS (FBNetV2-XLarge+MCA)	2.36	89.8	96.4
FBNetV2-XLarge	2.22	88.5	95.5
RIB-DCS-DNAS (FBNetV2-XLarge)	2.38	89.7	96.2

A.10 COMPARISON WITH EXISTING ATTENTION MASKING METHOD

In this section, we compare RIB-DCS with an existing attention masking method, MODA [Zhang et al., 2025], which is proposed to mitigate attention imbalance in multimodal Transformers by introducing a non-learnable modular attention mask. MODA constructs its mask by combining a predefined decay prior with pseudo-attention statistics extracted from intermediate activations, assigning progressively larger negative penalties to token pairs based on their positional distance in the token sequence. In the single-modality setting, such as person Re-ID, MODA can be applied by treating all visual patches as a single token stream. The same decay mechanism is used to regulate how strongly a patch is allowed to attend to patches that are sequentially distant. As a result, MODA produces a mask on the attention weights that enforces a token-level structural constraint that suppresses excessive long-range attention and promotes more balanced attention distributions.

Table 13: Comparison between RIB-DCS and existing attention masking method, MODA [Zhang et al., 2025], for person Re-ID on Market1501.

Model	FLOPs	mAP	Rank-1
TransReID	19.3 G	88.9	95.2
MODA (TransReID)	19.4 G	89.4	95.3
RIB-DCS-FB (TransReID)	19.7 G	91.3	97.0

However, such a positional decay prior may not be beneficial for person Re-ID, where discriminative visual patterns are often scattered across spatially distant body regions. For example, the head, clothes, backpack, and shoes of a person may occupy distant patches in the token sequence, while jointly providing essential identity-critical information. Penalizing interactions among these patches that are far apart risks weakening meaningful long-range dependencies and degrading the model’s ability to capture informative features related to the person’s identity. In contrast with MODA, our proposed RIB-DCS aims to learn an informative weight matrix that makes the learned features more correlated with the class labels and less correlated with the input features, without relying on unreliable priors. Instead of assigning penalties based on spatial proximity, RIB-DCS learns to remove uninformative channels while preserving those that carry identity-relevant information. The mask module is optimized jointly with the network parameters under the guidance of RIB. To demonstrate the advantages of RIB-DCS over MODA, we perform an experiment by applying MODA to all the attention modules in TransReID, leading to MODA (TransReID). It is observed in Table 13 that RIB-DCS-FB (TransReID) significantly outperforms MODA (TransReID) by 1.9% in mAP and 1.7% in Rank-1 for person Re-ID on Market1501.

A.11 EVALUATION ON CUHK03 DATASET

We further evaluate the effectiveness of the proposed RIB on the CUHK03 [Li et al., 2014] dataset. The labeled version of the dataset, where the bounding boxes of the persons are manually annotated by humans, is used following the configurations in [Li et al., 2014], without re-ranking or multi-query fusion. We compare the performance of RIB-DCS-FB (TransReID), RIB-DCS-DNAS (FBNetV2-XLarge), RIB-CA-FB (TransReID+MCA), and RIB-CA-DNAS (FBNetV2-XLarge+MCA) with their corresponding baseline models without RIB. It is observed in Table 14 that RIB significantly improves the performance of all the baseline models. For instance, RIB-DCS-FB (TransReID) outperforms TransReID by 1.6% in mAP.

Table 14: Performance comparison on the CUHK03 dataset under the traditional evaluation protocol. Performance improvements over baseline models are indicated in parentheses.

Method	mAP (%)	Rank-1 (%)
TransReID [He et al., 2021]	79.6	81.7
RIB-DCS-FB (TransReID)	81.2 (↑ 1.6)	82.7 (↑ 1.0)
FBNetV2-XLarge [Wan et al., 2020]	78.0	80.3
RIB-DCS-DNAS (FBNetV2-XLarge)	79.6 (↑ 1.6)	81.1 (↑ 0.8)
TransReID+MCA [He et al., 2021]	80.1	81.8
RIB-CA-FB (TransReID+MCA)	81.1 (↑ 1.0)	82.5 (↑ 0.7)
FBNetV2-XLarge+MCA [Wan et al., 2020]	78.4	80.6
RIB-CA-DNAS (FBNetV2-XLarge+MCA)	79.7 (↑ 1.3)	81.3 (↑ 0.7)

A.12 CROSS-DOMAIN PERSON RE-ID

To further evaluate the cross-domain generalization capability of the proposed RIB method, we integrate RIB-DCS into the domain transfer framework of CaCL [Lee et al., 2023], which performs contrastive adaptation across heterogeneous person Re-ID datasets. In particular, CaCL aims to bridge the domain gap between the source and target datasets through class-aware contrastive learning and domain-level feature alignment. Both our model, RIB-DCS-FB (CaCL), and the baseline model, CaCL, are evaluated under the standard cross-dataset Re-ID protocol [Lee et al., 2023], where training is performed on one dataset (source domain), and testing is conducted on another (target domain). It is observed in Table 15 that RIB-DCS-FB (CaCL) significantly outperforms the corresponding baseline model, CaCL. For instance, RIB-DCS-FB (CaCL) outperforms CaCL by 1.7% in mAP when the models are trained on Market1501 and tested on MSMT17, which demonstrates the capabilities of the RIB in learning informative features from the input image for identifying the person’s identity even when the test data is not from the same domain as the training data.

Table 15: Performance comparison for cross-domain evaluation on MSMT17 and Market1501. Performance improvements over the baseline model CaCL [Lee et al., 2023] are indicated in parentheses.

Method	Training	Test	mAP	Rank-1
CaCL [Lee et al., 2023]	MSMT17	Market1501	84.7	93.8
RIB-DCS-FB (CaCL)	MSMT17	Market1501	85.7 (↑ 1.0)	94.6 (↑ 0.8)
CaCL [Lee et al., 2023]	Market1501	MSMT17	66.6	36.5
RIB-DCS-FB (CaCL)	Market1501	MSMT17	68.3 (↑ 1.7)	37.7 (↑ 1.2)

A.13 INTEGRATION WITH SELF-SUPERVISED RE-IDENTIFICATION FRAMEWORKS

To further demonstrate the general applicability of the proposed RIB, we integrate RIB-DCS into three representative self-supervised person re-identification frameworks, including the CLIP-based method CLIP-REID [Li et al., 2023], the DINO-based method SOLIDER [Chen et al., 2023], and the MAE-based method PersonMAE [Hu et al., 2024]. These frameworks represent three distinct paradigms of self-supervised representation learning, including vision-language contrastive alignment, self-distillation via teacher-student consistency, and masked image reconstruction, respectively. Following the settings in PersonMAE [Hu et al., 2024], all models are pre-trained on LUPerson-4.2M [Fu et al., 2021]. For each framework, the DCS-Attention modules are incorporated into the backbone feature extractor to enable differentiable selection of informative channels in the computation of attention weights. The IB regularization, implemented via reducing the proposed IBB, is jointly optimized with the training loss for each of the self-supervised learning methods. It is observed in Table 16 that the RIB augmented models based on CLIP, DINO, and MAE consistently outperform the corresponding baseline models. For instance, RIB-DCS-FB (CLIP-REID) outperforms CLIP-REID by 1.2% in mAP for person Re-ID on Market1501, which demonstrates the general applicability of RIB in learning informative features within modern self-supervised feature learning frameworks.

Table 16: Performance comparison of self-supervised person Re-ID frameworks and their DCS-enhanced variants on Market1501. All feature backbones are pre-trained on LUPerson-4.2M [Fu et al., 2021].

Method	Backbone	mAP (%)	Rank-1 (%)
CLIP-REID [Li et al., 2023]	TransReID	89.6	95.5
RIB-DCS-FB (CLIP-REID)	RIB-DCS-FB (TransReID)	90.8	96.6
SOLIDER [Chen et al., 2023]	TransReID	92.5	96.0
RIB-DCS-FB (SOLIDER)	RIB-DCS-FB (TransReID)	93.8	97.2
PersonMAE [Hu et al., 2024]	TransReID	93.6	97.1
RIB-DCS-FB (PersonMAE)	RIB-DCS-FB (TransReID)	94.3	97.5

A.14 ABLATION STUDY ON THE EFFECTS OF DIFFERENTIABLE CHANNEL SELECTION FOR ATTENTION WEIGHTS

In this section, we study the effectiveness of the DCS-Attention modules in RIB-DCS-FB and RIB-DCS-DNAS models. The study is performed using three baseline models, including MobileNetV2, TransReID, and FBNetV2-Large. Since vanilla MobileNetV2 and FBNetV2-Large do not have self-attention modules, we build two baseline models that incorporate vanilla self-attention modules, leading to MobileNetV2+Self-Attention and FBNetV2-Large+Self-Attention. We also build ablation models that add IBB in the training loss on TransReID, MobileNetV2+Self-Attention, and FBNetV2-Large+Self-Attention. The IBB is incorporated into the above models with vanilla self-attention modules to encourage the attention outputs to be more informative than those from the models trained without IBB. The models with vanilla self-attention and IBB are denoted as MobileNetV2+Self-Attention+IBB, TransReID+IBB, and FBNetV2-Large+Self-Attention+IBB. It is observed in Table 17 that reducing IBB in the training of the vanilla models and the ablation models with vanilla self-attention improves their performance for person Re-ID, which demonstrates the effectiveness of IBB in promoting the learning of informative attention outputs. Moreover, the performance of the models with vanilla self-attention modules and IBB can be further improved by replacing the vanilla self-attention modules with the proposed DCS-Attention modules. In particular, our RIB-DCS-FB and RIB-DCS-DNAS models outperform the vanilla models and the ablation study models by large margins in terms of mAP and Rank-1. For example, RIB-DCS-FB (TransReID) outperforms the vanilla model, TransReID, by 2.4% in mAP. In addition, RIB-DCS-FB (TransReID) outperforms the ablation model TransReID+IBB by 1.1% in mAP, demonstrating the effectiveness of the DCS-Attention module by learning informative attention weights.

Table 17: Ablation study on the effects of DCS-Attention. The comparisons are performed on Market1501.

Model	FLOPs	mAP	Rank-1
MobileNetV2	0.382 G	84.5	93.9
MobileNetV2+IBB	0.382 G	85.2	94.6
MobileNetV2+Self-Attention+IBB	0.395 G	85.6	94.6
RIB-DCS-FB (MobileNetV2)	0.406 G	86.4	95.1
TransReID	19.3 G	88.9	95.2
TransReID+IBB	19.3 G	90.2	96.0
RIB-DCS-FB (TransReID)	19.7 G	91.3	97.0
FBNetV2-Large	1.235 G	87.2	95.0
FBNetV2-Large+IBB	1.235 G	87.8	95.5
FBNetV2-Large+Self-Attention+IBB	1.288 G	88.4	95.5
RIB-DCS-DNAS (FBNetV2-Large)	1.302 G	89.1	95.8

A.15 VISUALIZATION ANALYSIS

In this section, we apply Grad-CAM [Selvaraju et al., 2017] to visualize the effectiveness of RIB-DCS and RIB-CA models for person Re-ID. We use Grad-CAM to visualize the regions within the input images that influence the predictions of different models.

The Grad-CAM visualization in Fig. 3(c) in Section A.15 illustrates that representative baselines in self-attention, TransReID [He et al., 2021], and channel attention, SE [Hu et al., 2018], either focus on background areas or miss important human body regions, potentially degrading person Re-ID performance. Fig. 4 illustrates the Grad-CAM visualization results for the vanilla TransReID, RIB-DCS-FB (TransReID)+DCS-Attention, and our RIB-DCS-FB (TransReID). The Grad-CAM visualization results illustrate that the representative self-attention modules in TransReID usually miss the important parts of human figures, potentially hurting the performance in person Re-ID. In contrast, RIB-DCS-FB focuses more strongly on the important parts of human figures in the input that are informative for person Re-ID. More Grad-CAM visualization results for RIB-DCS-FB (MobileNetV2) and the corresponding baseline models are illustrated in Fig. 5.

Fig. 6 illustrates that the baseline model, TransReID, either mistakenly focuses on the background areas or misses the important parts of human figures affected by pose changes. In contrast, RIB-DCS-FB (TransReID), mostly highlights informative and identity-related body parts across both poses. The additional Grad-CAM visualization results demonstrate that the proposed RIB method promotes the learning of more robust and informative representations under challenging conditions, such as occlusion and extreme pose variations.

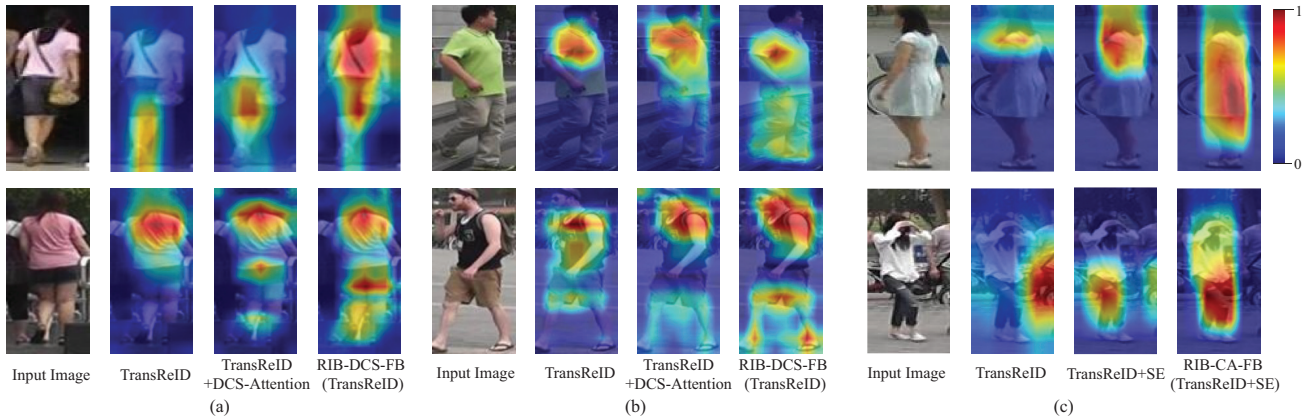


Figure 3: Figures (a)-(b) illustrate the Grad-CAM heatmaps for TransReID, TransReID+DCS-Attention, and RIB-DCS-FB (TransReID). TransReID+DCS-Attention replaces the self-attention modules in TransRe-ID with the DCS-Attention modules. Figure (c) illustrates the Grad-CAM heatmaps for TransReID, TransReID+SE, and RIB-CA-FB (TransReID+SE). The SE attention modules [Hu et al., 2018] are inserted after each transformer block in TransReID+SE.

Fig. 7 illustrate the Grad-CAM heatmaps for TransReID, TransReID+SE, TransReID+CBAM, RIB-CA-FB (TransReID+SE), and RIB-CA-FB (TransReID+CBAM). The baseline models, TransReID+SE and TransReID+CBAM, incorporate the SE [Hu et al., 2018] attention modules and the CBAM [Woo et al., 2018] attention modules by inserting them after each transformer block in the backbone of TransReID. RIB-CA-FB (TransReID+SE) and RIB-CA-FB (TransReID+CBAM) apply RIB on the backbones of TransReID+SE and TransReID+CBAM. The Grad-CAM visualization results illustrate that the representative channel attention modules, SE and CBAM, either mistakenly focus on the background areas or miss the important human body regions, potentially hurting the performance of person Re-ID. In contrast, RIB-CA-FB focuses more strongly on the person’s body in the input image.

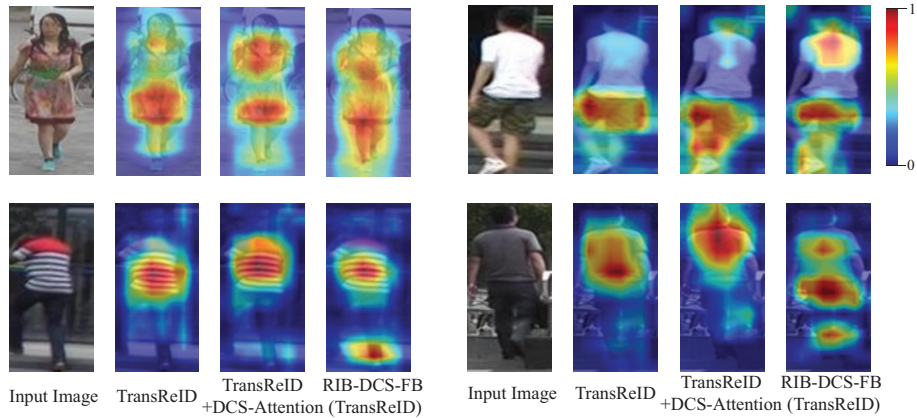


Figure 4: Grad-CAM visualization results for TransReID, TransReID+DCS-Attention, and RIB-DCS-FB (TransReID). TransReID+DCS-Attention replaces all the self-attention modules in TransReID with the DCS-Attention modules.

A.16 T-SNE VISUALIZATION OF LEARNED FEATURES

To further assess the effectiveness of RIB in learning discriminative features, we perform the t-SNE visualization analysis on features learned by RIB-DCS-FB (TransReID) and the baseline model, TransReID. The visualization is conducted using images of 10 randomly selected identities from the Market-1501 dataset, where each identity has 20 images. Each identity is considered a distinct class. The features are extracted from the output of the last transformer block before the classification head in the TransReID and RIB-DCS-FB (TransReID) models. Standard normalization, which standardizes features to have zero mean and unit variance, is applied to the learned features for both TransReID and RIB-DCS-FB (TransReID) to ensure

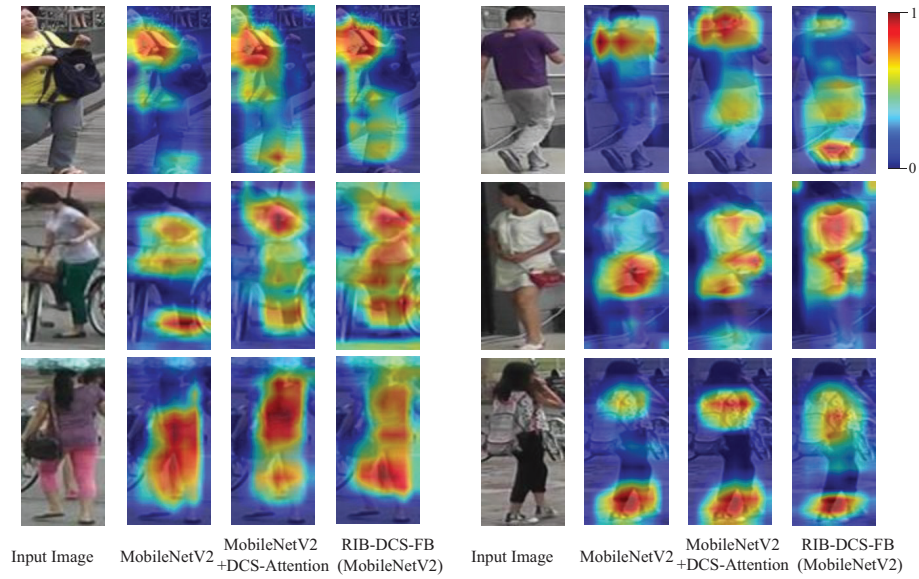


Figure 5: Grad-CAM visualization results for MobileNetV2, MobileNetV2+DCS-Attention, and RIB-DCS-FB (MobileNetV2). MobileNetV2+DCS-Attention inserts the DCS-Attention module after each of its residual blocks.

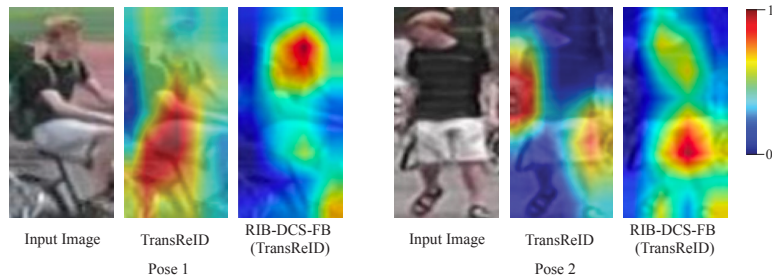


Figure 6: Grad-CAM visualization of a person under extreme pose variation in Markert1501 for TransReID and RIB-DCS-FB (TransReID).

consistent scaling across dimensions. To prepare the features for discriminative analysis, we perform Linear Discriminant Analysis (LDA), which projects the original high-dimensional features into a lower-dimensional subspace that maximizes class separability [Fisher, 1936], on the features generated by both TransReID and RIB-DCS-FB (TransReID). The number of LDA components, which is the dimension of the LDA subspace, is set to 9. Next, we perform t-SNE visualization on the LDA-projected features for both TransReID and RIB-DCS-FB (TransReID) to qualitatively assess the discriminative capacity of the learned representations. As illustrated in Fig. 8, the features learned by RIB-DCS-FB (TransReID) exhibit better inter-class separability and increased intra-class compactness than those obtained without RIB-DCS-FB. For example, features generated by TransReID for instances from five distinct classes exhibit substantial overlap within the red dashed box in Fig. 8, which indicates limited inter-class separability and reduced intra-class compactness. In contrast, the features generated by RIB-DCS-FB (TransReID) for instances from different classes are more distinctly clustered and spatially separated from one another. This qualitative observation underscores the effectiveness of the informative channel selection in the proposed DCS-Attention in enhancing the discriminative capacity of the learned features. To further support this observation, we compute the Fisher Score [Duda et al., 2001] of the LDA-projected features, which quantitatively evaluates the discriminative capability of the features by measuring the ratio of inter-class variance to intra-class variance. The Fisher Score of the LDA-projected features of RIB-DCS-FB (TransReID) is 7.173, which is 121% higher than that of TransReID, which is 3.246. This result quantitatively confirms that the proposed DCS-Attention enhances the feature discriminability, aligning well with the patterns observed in the t-SNE visualization.

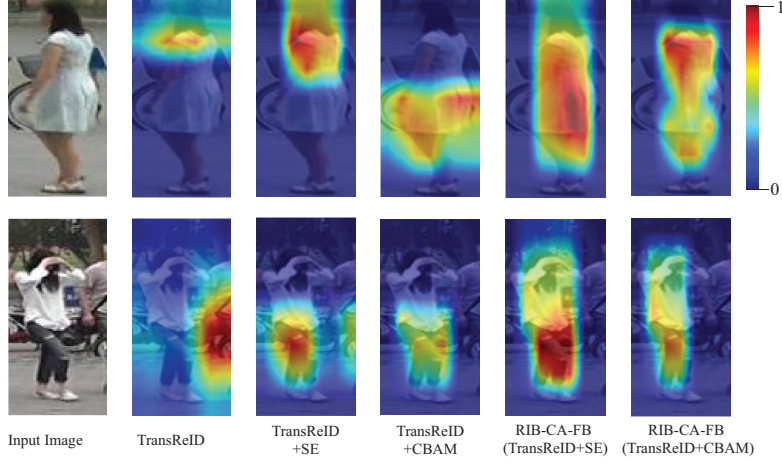


Figure 7: Grad-CAM visualization results for TransReID, TransReID+SE, TransReID+CBAM, RIB-CA-FB (TransReID+SE), and RIB-CA-FB (TransReID+CBAM). The SE attention modules and CBAM attention modules are inserted after each transformer block in TransReID+SE and TransReID+CBAM, respectively.

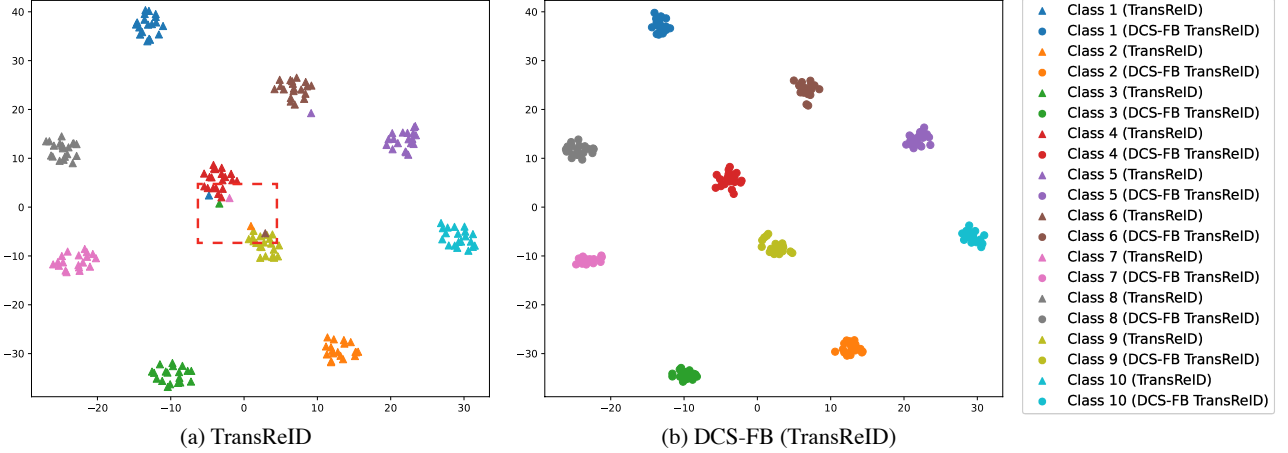


Figure 8: t-SNE visualization analysis of the features learned by (a) TransReID and (b) RIB-DCS-FB (TransReID). The visualization is performed on the LDA-projected features for better discriminative analysis. The red dashed box in Figure (a) shows an example where features generated by TransReID for instances from multiple distinct classes exhibit substantial overlap.

A.17 COMPARISON WITH RECENT STATE-OF-THE-ART PERSON RE-ID METHODS

In this section, we further evaluate the effectiveness of RIB on top of recent state-of-the-art person Re-ID methods to better position our contribution with respect to the current state of the art. We emphasize that the proposed RIB has already been applied to baseline methods published in 2024, including SPT [Tan et al., 2024] in Section 4.5 and MCA [Jiang et al., 2024] in Section 4.3, where RIB significantly improves the performance of both SPT and MCA. To further compare with more recent baselines, we apply RIB-DCS to two recent state-of-the-art methods, PersonViT [Hu et al., 2025] (MVA 2025) and HPL [Zhou et al., 2026] (AAAI 2026). Following the naming convention of the RIB-DCS-FB models in the main paper, the resulting models are denoted as RIB-DCS-FB (PersonViT) and RIB-DCS-FB (HPL). RIB-DCS is applied by replacing the standard self-attention in PersonViT and HPL with the DCS-Attention modules and incorporating IBB into the training loss, while following the original training settings of PersonViT [Hu et al., 2025] and HPL [Zhou et al., 2026]. It is observed in Table 18 that RIB-DCS consistently improves both recent baselines across Market1501, DukeMTMC, and MSMT17. For example, RIB-DCS-FB (HPL) outperforms HPL by 1.4% in mAP on MSMT17. Moreover, RIB-DCS-FB (PersonViT) achieves 96.2% mAP on Market1501, outperforming PersonViT by 1.2% in mAP, further demonstrating that the proposed

IBB framework remains competitive with the current state-of-the-art Re-ID methods.

Table 18: Performance comparison between recent state-of-the-art person Re-ID methods, PersonViT [Hu et al., 2025] (MVA 2025) and HPL [Zhou et al., 2026] (AAAI 2026), and their RIB-DCS-FB variants on Market1501, DukeMTMC, and MSMT17. The improvements of the RIB models over their corresponding baselines are reported in parentheses after each metric value.

Method	Backbone	Market1501		DukeMTMC		MSMT17	
		mAP	Rank-1	mAP	Rank-1	mAP	Rank-1
PersonViT [Hu et al., 2025]	ViT-B/16	95.0	97.6	88.1	93.8	80.8	92.0
RIB-DCS-FB (PersonViT)	ViT-B/16	96.2 ($\uparrow$ 1.2)	98.3 ($\uparrow$ 0.7)	89.2 ($\uparrow$ 1.1)	94.7 ($\uparrow$ 0.9)	82.0 ($\uparrow$ 1.2)	93.0 ($\uparrow$ 1.0)
HPL [Zhou et al., 2026]	ViT-B/16	89.8	96.0	82.9	90.4	79.0	91.0
RIB-DCS-FB (HPL)	ViT-B/16	91.1 ($\uparrow$ 1.3)	97.1 ($\uparrow$ 1.1)	84.2 ($\uparrow$ 1.3)	91.5 ($\uparrow$ 1.1)	80.4 ($\uparrow$ 1.4)	92.2 ($\uparrow$ 1.2)

A.18 EXTENSION OF RIB TO PROTOTYPICAL LEARNING (RIBPL) AND SENSITIVITY TO THE NUMBER OF PROTOTYPES

The class-centroid soft assignment in RIB, described in Section 3.1, fixes the number of centroids to the number of classes C . To further improve the modeling capability over that of class centroids, we extend the class-centroid-based learning of RIB to prototypical learning (PL) [Caron et al., 2020, Li et al., 2021a, Yang et al., 2018, Snell et al., 2017], where we use prototypes, that is, the clustering centroids, instead of class centroids. The resulting models are termed RIBPL. Following the general framework of RIB in Section 3.2, RIBPL can be applied to both DCS-Attention and existing channel attention modules, leading to RIBPL-DCS and RIBPL-CA. The number of prototypes is larger than the number of class centroids C , and is selected by standard cross-validation [Caron et al., 2020, Li et al., 2021a] from $2C$ to $20C$. With more prototypes than classes, the distribution of the latent feature space is better captured [Caron et al., 2020, Li et al., 2021a]. Consequently, the probability density modeled by PL is closer to the ground-truth continuous feature density, making the IBB bound tighter. The corresponding theoretical error bound is the gap between the prototype-based density estimation and the ground-truth density, which can be bounded using standard statistical analyses such as kernel density estimation (KDE) [Kim and Scott, 2012].

We first evaluate RIBPL-DCS and RIBPL-CA using TransReID [He et al., 2021] as the feature backbone, where RIBPL-CA is built on the best channel attention module, MCA [Jiang et al., 2024]. As shown in Table 19, both RIBPL-DCS-FB (TransReID) and RIBPL-CA-FB (TransReID+MCA) further improve their class-centroid counterparts, RIB-DCS-FB (TransReID) and RIB-CA-FB (TransReID+MCA), across all three datasets. We further conduct a sensitivity study by varying the number of prototypes in $\{2C, 5C, 10C, 15C, 20C\}$. As shown in Table 19, the performance of both RIBPL-DCS and RIBPL-CA remains stable with different numbers of prototypes. We remark that both the original RIB models and the RIBPL models use the 10-epoch warm-up stage described in Section A.2, where only the cross-entropy and triplet losses are optimized. This ensures that the latent features are sufficiently informative before the class centroids or prototypes are computed. The centroids/prototypes are then initialized as averages of the warmed-up features, so that they are implicitly regularized by a moderately trained network, which stabilizes the training. This warm-up strategy is also widely used in the prototypical learning literature [Caron et al., 2020, Li et al., 2021a].

We also evaluate RIBPL-DCS using HRNet [Dou et al., 2022] as the fixed feature extraction backbone, since RIB-DCS-FB (HRNet) renders relatively modest gains over HRNet in the main paper. As shown in Table 20, RIBPL-DCS-FB (HRNet) further improves the performance of RIB-DCS-FB (HRNet) and significantly outperforms the baseline HRNet. For example, RIBPL-DCS-FB (HRNet) outperforms HRNet by 1.1% in mAP on DukeMTMC.

A.19 QUANTITATIVE EVALUATION OF ATTENTION QUALITY AND THE CONTROLLED EFFECT OF IBB

To further enhance the evidence that reducing IBB leads to more informative attention beyond the qualitative Grad-CAM visualizations, we perform a quantitative evaluation of the attention quality of RIB-DCS. Motivated by recent analyses of semantic query-key interactions in Vision Transformers, where foreground/background interactions are used to interpret attention behavior [Pan et al., 2024], we evaluate whether foreground person tokens attend more to other foreground person tokens rather than to background regions. We also report an attention-overlap metric inspired by Attention IoU [Serianni et al., 2025], which measures the spatial agreement between the Grad-CAM activation map and the foreground person mask.

Table 19: Sensitivity study of RIBPL-DCS-FB (TransReID) and RIBPL-CA-FB (TransReID+MCA) with respect to the number of prototypes on Market1501, DukeMTMC, and MSMT17. C denotes the number of classes (identities), and the rows with C prototypes correspond to the class-centroid models RIB-DCS-FB (TransReID) and RIB-CA-FB (TransReID+MCA).

Method	#Prototypes	Market1501		DukeMTMC		MSMT17	
		mAP	Rank-1	mAP	Rank-1	mAP	Rank-1
RIB-DCS-FB (TransReID)	C	91.3	97.0	84.3	92.0	70.2	87.1
RIBPL-DCS-FB (TransReID)	$2C$	91.8	97.4	84.8	92.5	70.7	87.6
RIBPL-DCS-FB (TransReID)	$5C$	91.9	97.5	84.9	92.6	70.8	87.7
RIBPL-DCS-FB (TransReID)	$10C$	91.7	97.3	84.7	92.4	70.6	87.5
RIBPL-DCS-FB (TransReID)	$15C$	91.6	97.2	84.6	92.3	70.5	87.4
RIBPL-DCS-FB (TransReID)	$20C$	91.5	97.2	84.5	92.2	70.4	87.5
RIB-CA-FB (TransReID+MCA)	C	91.0	96.8	84.1	91.9	69.5	86.8
RIBPL-CA-FB (TransReID+MCA)	$2C$	91.4	97.1	84.5	92.3	70.0	87.2
RIBPL-CA-FB (TransReID+MCA)	$5C$	91.5	97.2	84.6	92.4	70.1	87.2
RIBPL-CA-FB (TransReID+MCA)	$10C$	91.3	97.0	84.4	92.2	69.9	87.1
RIBPL-CA-FB (TransReID+MCA)	$15C$	91.2	96.9	84.3	92.1	69.8	87.0
RIBPL-CA-FB (TransReID+MCA)	$20C$	91.1	96.9	84.2	92.0	69.7	87.0

Table 20: Performance of RIBPL-DCS-FB (HRNet) compared with RIB-DCS-FB (HRNet) and the baseline HRNet on Market1501, DukeMTMC, and MSMT17.

Method	Params(M)	FLOPs(G)	Market1501		DukeMTMC		MSMT17	
			mAP	Rank-1	mAP	Rank-1	mAP	Rank-1
HRNet [Dou et al., 2022]	30.5	9.0	89.5	95.7	82.5	90.8	65.3	84.7
RIB-DCS-FB (HRNet)	32.1	9.9	90.1	96.2	83.0	91.1	66.9	85.1
RIBPL-DCS-FB (HRNet)	32.1	9.9	90.7	96.7	83.6	91.8	67.1	85.8

For each image, we first obtain a foreground person mask using an off-the-shelf segmentation model, DINO [Caron et al., 2021], and downsample it to the token grid following [Pan et al., 2024]. Let $A \in \mathbb{R}^{N \times N}$ denote the self-attention weight matrix over N image tokens from the last transformer block of the model, and let $m \in (0, 1)^N$ be the foreground mask obtained by DINO. We define the foreground-foreground attention ratio (FG-FG) and the foreground-background attention ratio (FG-BG) as

$$\text{FG-FG} = \frac{m^\top A m}{\sum_{i=1}^N m_i}, \quad \text{FG-BG} = \frac{m^\top A (1 - m)}{\sum_{i=1}^N m_i},$$

which measure the ratios of attention weights from foreground query tokens to foreground and to background key tokens, respectively. We further compute Attention-IoU by measuring the spatial overlap between the foreground person mask and the binarized Grad-CAM activation map following [Serianni et al., 2025]. Let $g \in [0, 1]^N$ denote the token-level Grad-CAM activation map normalized to $[0, 1]$, and let $\hat{g} \in \{0, 1\}^N$ be its binarized version with $\hat{g}_i = \mathbb{I}_{\{g_i > 0.5\}}$. The Attention-IoU is then defined as

$$\text{Attention-IoU} = \frac{\sum_{i=1}^N \hat{g}_i m_i}{\sum_{i=1}^N \mathbb{I}_{\{\hat{g}_i + m_i > 0\}}}.$$

A higher FG-FG value, a lower FG-BG value, and a higher Attention-IoU indicate that the attention and the learned representation focus more on identity-relevant person regions and less on background, which are preferred for person Re-ID. We perform the evaluation using RIB-DCS-FB (TransReID) with different balance factors $\eta \in \{0.1, 0.5, 1.0, 1.5, 2.0\}$ for IBB and compare it with the baseline model TransReID [He et al., 2021] on Market1501, DukeMTMC, and MSMT17.

The results in Table 21 show that, compared with TransReID, RIB-DCS-FB consistently increases FG-FG and Attention-IoU while reducing FG-BG, indicating that the learned attention and representation are more concentrated on identity-relevant regions that are more informative for person Re-ID. Table 21 also presents a controlled experiment demonstrating how the attention distributions change as the IBB term is varied with different balance factors η . It is observed that increasing η from 0.1 to 1.0 consistently improves the attention quality across all three datasets, and the performance remains stable when η is further increased to 1.5 or 2.0, which is consistent with the choice of $\eta = 1$ selected by cross-validation in Section A.3.

We further clarify that the link between IBB and informative attention is not based solely on visualization, but follows from the role of attention in the information flow from the input X to the learned representation F . In RIB, the attention module lies on the computational path from X to F . For RIB-DCS, the attention weights are computed by $A_\theta = \sigma((X \odot \mathcal{M}_\theta(X))(X \odot \mathcal{M}_\theta(X))^\top)$, where the differentiable mask $\mathcal{M}_\theta$ is jointly optimized with the network. Hence, the IBB loss directly updates the attention parameters through

$$\nabla_\theta \text{IBB} = \frac{\partial \text{IBB}}{\partial F} \frac{\partial F}{\partial A_\theta} \frac{\partial A_\theta}{\partial \mathcal{M}_\theta} \frac{\partial \mathcal{M}_\theta}{\partial \theta}.$$

Similarly, the IBB loss directly updates the attention parameters of the channel attention modules in RIB-CA. Since IBB upper-bounds the IB loss $I(F, X) - I(F, Y)$, reducing IBB encourages F to preserve identity-relevant information while suppressing nuisance information from X , such as occlusions and background. This optimization principle is consistent with prior IB-based attention studies [Zhou et al., 2022, Lai et al., 2021]. Therefore, reducing IBB establishes a direct optimization-based mechanism for learning informative attention, while the Grad-CAM and attention visualizations together with the quantitative evaluation in Table 21 serve as direct supporting evidence.

Table 21: Quantitative evaluation of attention quality of RIB-DCS-FB (TransReID) with different balance factors η for IBB, compared with the baseline TransReID [He et al., 2021], on Market1501, DukeMTMC, and MSMT17. A higher FG-FG value, a lower FG-BG value, and a higher Attention-IoU indicate that the attention and the learned representation focus more on identity-relevant foreground person regions.

Dataset	Method	η	FG-FG	FG-BG	Attention-IoU
Market1501	TransReID	–	70.4	29.6	51.4
	RIB-DCS-FB (TransReID)	0.1	73.8	26.2	54.7
	RIB-DCS-FB (TransReID)	0.5	75.2	24.8	55.9
	RIB-DCS-FB (TransReID)	1.0	76.1	24.0	56.8
	RIB-DCS-FB (TransReID)	1.5	76.0	23.9	56.7
	RIB-DCS-FB (TransReID)	2.0	76.1	24.0	56.6
DukeMTMC	TransReID	–	68.2	31.8	48.6
	RIB-DCS-FB (TransReID)	0.1	71.7	28.3	51.6
	RIB-DCS-FB (TransReID)	0.5	73.2	26.8	53.0
	RIB-DCS-FB (TransReID)	1.0	74.1	25.9	53.9
	RIB-DCS-FB (TransReID)	1.5	73.9	26.1	53.9
	RIB-DCS-FB (TransReID)	2.0	73.9	26.0	53.9
MSMT17	TransReID	–	63.3	36.7	43.2
	RIB-DCS-FB (TransReID)	0.1	66.8	33.2	46.5
	RIB-DCS-FB (TransReID)	0.5	68.7	31.3	48.2
	RIB-DCS-FB (TransReID)	1.0	69.9	30.1	49.0
	RIB-DCS-FB (TransReID)	1.5	69.7	30.3	49.1
	RIB-DCS-FB (TransReID)	2.0	69.6	30.4	48.8

A.20 ABLATION STUDY ON THE CONTRIBUTION OF EACH LOSS TERM

To study the contribution of each loss term to the overall performance, we conduct an ablation study on the training objective of RIB models. The study is performed on RIB-DCS-FB with TransReID [He et al., 2021] as the backbone. The full objective in Equation (2) combines the cross-entropy (CE) loss, the triplet loss, and the proposed IBB. We compare three variants: IBB only, CE loss + triplet loss, and CE loss + triplet loss + IBB. The results in Table 22 show that the best performance is achieved when IBB is combined with the CE loss and the triplet loss. For example, adding IBB to the CE loss and the triplet loss improves the mAP by 1.6% and the Rank-1 accuracy by 1.0% on Market1501, demonstrating the effectiveness of IBB in regularizing the representation learned by the CE loss and the triplet loss. Using IBB alone is not sufficient to achieve as good performance as using the CE loss and the triplet loss. This is because IBB is not intended to replace the standard metric learning losses, but to serve as a complementary information-theoretic regularizer. The CE loss and the triplet loss provide strong supervised identity discrimination and metric structure, and IBB further regularizes the learned representation by encouraging the model to learn informative attention weights that retain identity-relevant information for person Re-ID.

Table 22: Ablation study on the contribution of each loss term in the training objective of RIB-DCS-FB (TransReID) on Market1501, DukeMTMC, and MSMT17.

Objective	Market1501		DukeMTMC		MSMT17	
	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1
IBB	89.5	95.7	82.2	90.4	67.6	85.1
CE + Triplet	89.7	96.0	83.0	91.0	68.9	85.8
CE + Triplet + IBB	91.3	97.0	84.3	92.0	70.2	87.1

A.21 ABLATION STUDY ON THE INDIVIDUAL CONTRIBUTIONS OF DCS-ATTENTION AND IBB

To isolate the individual contributions of the DCS-Attention module and IBB, we combine the corresponding ablation results, which are also related to Table 7 in Section A.4 and Table 17 in Section A.14, into Table 23 for convenience. The comparisons are performed on Market1501 using MobileNetV2 as the feature backbone. It is observed that both the DCS-Attention module and IBB individually improve the baseline MobileNetV2, demonstrating the individual contribution of each component. In addition, the proposed RIB-DCS-FB (MobileNetV2), which combines DCS-Attention and IBB for learning informative attention, achieves the best performance.

Table 23: Ablation study on the individual contributions of the DCS-Attention module and IBB. The comparisons are performed on Market1501 using MobileNetV2 as the feature backbone.

Model	mAP	Rank-1
MobileNetV2	84.5	93.9
MobileNetV2+DCS-Attention	85.8	94.5
MobileNetV2+IBB	85.2	94.6
RIB-DCS-FB (MobileNetV2)	86.4	95.1

Table 24: Training time, search time, and inference time of RIB models and their corresponding baseline models on Market1501. The evaluation is conducted on one NVIDIA A100 GPU. The inference time (ms/img) is measured in inference mode with gradients disabled, using input images of size 256×128 and a batch size of 256.

Model	Training Time (h)	Search Time (h)	Inference Time (ms/img)	mAP	Rank-1
MobileNetV2	10.3	–	1.18	84.5	93.9
RIB-DCS-FB (MobileNetV2)	10.9	–	1.26	86.4	95.1
MobileNetV2+MCA	10.6	–	1.24	85.2	94.5
RIB-CA-FB (MobileNetV2+MCA)	11.2	–	1.31	86.1	95.0
TransReID	31.1	–	4.62	88.9	95.2
RIB-DCS-FB (TransReID)	32.9	–	4.74	91.3	97.0
TransReID+MCA	31.9	–	4.79	90.1	96.0
RIB-CA-FB (TransReID+MCA)	33.6	–	4.91	91.0	96.8
FBNetV2-Large	16.7	2.15	1.42	87.2	95.0
RIB-DCS-DNAS (FBNetV2-Large)	17.5	2.25	1.51	89.1	95.8
FBNetV2-XLarge+MCA	17.9	2.26	1.86	87.9	95.3
RIB-CA-DNAS (FBNetV2-XLarge+MCA)	18.7	2.36	1.95	89.8	96.4

A.22 INFERENCE TIME EVALUATION

In addition to the training time and search time evaluations in Sections A.8 and A.9, we further evaluate the inference time of the RIB-DCS models, the RIB-CA models, and the corresponding baseline models to demonstrate that RIB models do not introduce large inference overhead. The inference time evaluation is conducted on one NVIDIA A100 GPU using input images of size 256×128 . All models are evaluated in inference mode with gradients disabled, and the inference is performed using a batch size of 256 on Market1501. The inference time, together with the training and search time reported in the main paper, is summarized in Table 24. As shown in Table 24, both RIB-DCS and RIB-CA only marginally increase

the training time and inference time of the corresponding baseline models while significantly improving the performance for person Re-ID. For example, RIB-DCS-FB (TransReID) only marginally increases the training time and inference time of TransReID by 5.7% and 2.6%, respectively, while significantly improving the mAP by 2.4% on Market1501.

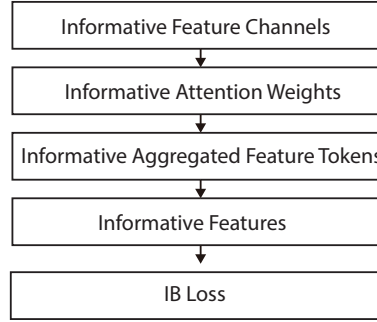


Figure 9: Illustration of the connection between the channel selection in DCS-Attention and the IB loss.

B ILLUSTRATION OF THE CONNECTION BETWEEN THE CHANNEL SELECTION IN DCS-ATTENTION AND THE REDUCTION OF THE IB LOSS

Fig. 9 illustrates the connection between the channel selection and the IB loss. In our DCS-Attention module, we propose to select informative feature channels so that more informative channels are used to learn more informative attention weights (please refer to our response to Question 1). By learning informative attention weights, DNNs assign higher attention weights to more informative feature tokens which contribute more to the discriminative task of person Re-ID. As a result, the aggregated feature tokens, which are the output of the DCS-Attention modules, are more informative for the discriminative task of person Re-ID than the aggregated feature tokens from vanilla self-attention modules. By replacing all the self-attention modules in DNNs with the DCS-Attention modules, the features learned by the DNNs are more informative for the discriminative task of person Re-ID. In order to reduce the IB loss, we propose an Information Bottleneck (IB) inspired channel selection for the attention weights, where the learned attention weights can be more informative by explicitly optimizing the IB loss for DNNs with DCS-Attention modules.

C STUDY ON THE VARIATIONAL UPPER BOUND FOR THE IB LOSS (IBB) AND TEST LOSS

Fig. 10 illustrates the test loss and the IBB during the training for RIB-DCS-FB (TransReID) and RIB-DCS-DNAS (FBNetV2). It is observed that the IBB for all the models decreases starting from the beginning of the training, and the test loss of a model trained with the IBB drops faster than that of the vanilla model.

D COMPUTATIONAL COMPLEXITY ANALYSIS

Given samples $\{(x_i, z_i)\}_{i=1}^n$ drawn from the joint distribution $p_\sigma(x, z) = p_\sigma(z|x)p(x)$, the goal of CLUB is to optimize the parameters of the predictive neural network $p_\sigma(z|x)$ such that the resulting joint model $p_\sigma(x, z)$ induces minimal mutual information between x and z . Here, $p_\sigma(z|x)$ serves as the main predictive model, such as an encoding model or a classification model, while $p(x)$ denotes the empirical data distribution. In addition, $q_\theta(z|x)$ is a variational conditional distribution, also implemented as a neural network, used to approximate $p(z|x)$ and to provide a differentiable estimator of mutual information. The optimization alternates between updating $q_\theta(z|x)$ by maximizing the log-likelihood $\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n \log q_\theta(z_i|x_i)$ to improve the approximation accuracy, and updating $p_\sigma(x, z)$ by minimizing the upper bound for the mutual information, $I(x, z)$, $\hat{I}_{\text{vCLUB}} = \frac{1}{n} \sum_{i=1}^n U_i$, where $U_i = \log q_\theta(z_i|x_i) - \frac{1}{n} \sum_{j=1}^n \log q_\theta(z_j|x_i)$. Through this alternating process, the algorithm jointly learns an accurate variational estimator q_θ and a predictive model p_σ . The complete training procedure is summarized in Algorithm 2.

Suppose the computation of $q_\theta(z|x)$ costs T_q and that of $p_\sigma(z|x)$ costs T_p . Computing $\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n \log q_\theta(z_i|x_i)$

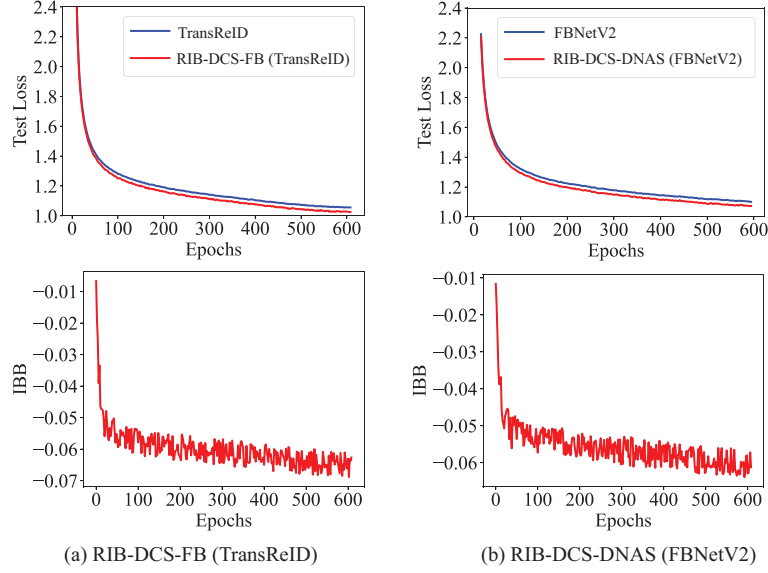


Figure 10: IBB and test loss of RIB-DCS-FB (TransReID) and RIB-DCS-DNAS (FBNetV2) during the training.

requires $\Theta(nT_q)$ time. The computation of $\text{vCLUB } \hat{I}_{\text{vCLUB}} = \frac{1}{n} \sum_{i=1}^n \left[\log q_\theta(z_i|x_i) - \frac{1}{n} \sum_{j=1}^n \log q_\theta(z_j|x_i) \right]$ requires computing $[\log q_\theta(z_j|x_i)]$ for $i, j \in [n]$, which requires $\Theta(n^2T_q)$ time. Updating p_σ for minimizing $\hat{I}_{\text{vCLUB}}$ over the same n inputs adds nT_p time. Hence, one training epoch requires $\Theta(n^2T_q + nT_q + nT_p) = \Theta(n^2T_q + nT_p)$.

Algorithm 2 MI Minimization with vCLUB (Algorithm 1 in CLUB [Cheng et al., 2020])

- 1: **for** each training iteration **do**
 - 2: Sample $\{(x_i, z_i)\}_{i=1}^n$ from $p_\sigma(x, z)$
 - 3: Compute log-likelihood $\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n \log q_\theta(z_i|x_i)$
 - 4: Update $q_\theta(z|x)$ by maximizing $\mathcal{L}(\theta)$
 - 5: **for** $i = 1$ to n **do**
 - 6: $U_i = \log q_\theta(z_i|x_i) - \frac{1}{n} \sum_{j=1}^n \log q_\theta(z_j|x_i)$
 - 7: **end for**
 - 8: Update $p_\sigma(x, z)$ by minimizing $\hat{I}_{\text{vCLUB}} = \frac{1}{n} \sum_{i=1}^n U_i$
 - 9: **end for**
-

Computational Complexity of IBB. Herein we analyze the computational complexity for computing the upper bound $\text{IBB}(\mathcal{W}) = \frac{1}{n} \sum_{i=1}^n \sum_{a=1}^C \sum_{y=1}^C \mathbb{I}_{\{y_i=y\}} \phi(F_i, a) \log \left(\frac{\mathbb{I}_{\{y_i=y\}}}{p_y Q(F \in a | Y=y)} \right)$ for the IB loss $I(F(\mathcal{W}), X)$ for a comparison with CLUB [Cheng et al., 2020]. Let T_0 denote the complexity of a forward and backward computation of the model predicting $F_i(\mathcal{W})$ for one sample. For each epoch, the computation of the soft assignments $\phi(F_i, a)$ for $i \in [n]$ and $a \in [C]$ requires $\Theta(nCT_0)$ time.

Once the soft assignment matrix $\phi(F_i, a)$ for all $i \in [n]$ and $a \in [C]$ has been obtained at the t -th epoch, the conditional distribution matrix $Q^{(t)}(F \in a | Y = y) \in \mathbb{R}^{C \times C}$ can be efficiently computed following Algorithm 3. We denote by $Q[a, y] = Q^{(t)}(F \in a | Y = y)$ the (a, y) -th entry of Q , representing the conditional probability that a feature F belongs to cluster a given class label y . Each entry $Q[a, y]$ is computed by aggregating the soft assignment values $\phi(F_i, a)$ over all $i \in [n]$ whose class label $y_i = y$, followed by normalization with respect to the total number of samples in that class. The accumulation step (lines 2–7 in Algorithm 3) requires $\Theta(nC)$ time, while the normalization step (lines 8–12 in Algorithm 3) requires $\Theta(C^2)$ time. Consequently, the overall computational complexity of computing Q is $\Theta(nC + C^2)$.

Let $\mathcal{I}_y = \{i \in [n] | y_i = y\}$ be the index set of the training samples from the class y for $y \in [C]$. Then, $\text{IBB}(\mathcal{W}) = \frac{1}{n} \sum_{i=1}^n \sum_{a=1}^C \sum_{y=1}^C \mathbb{I}_{\{y_i=y\}} \phi(F_i, a) \log \left(\frac{\mathbb{I}_{\{y_i=y\}}}{p_y Q(F \in a | Y=y)} \right) = \frac{1}{n} \sum_{a=1}^C \sum_{y=1}^C \sum_{i \in \mathcal{I}_y} \phi(F_i, a) \log \left(\frac{1}{p_y Q(F \in a | Y=y)} \right)$. As a result, the computation of $\text{IBB}(\mathcal{W})$ given $\phi(F_i, a)$ for $i \in [n]$ and $a \in [C]$ and Q takes $C \times \sum_{y=1}^C |\mathcal{I}_y| = \Theta(nC)$ time because $\sum_{y=1}^C |\mathcal{I}_y| = n$. The total computation cost for $\text{IBB}(\mathcal{W})$ is $\Theta(nCT_0 + nC + C^2 + nC) = \Theta(nCT_0 + C^2)$.

Algorithm 3 Efficient Computation of $Q^{(t)}(F \in a \mid Y = y)$

Input: The precomputed soft assignments $\phi(F_i, a)$ for $i \in [n]$ and $a \in [C]$, the class labels $\{y_i\}_{i=1}^n$, and the number of clusters C . Here $\{F_i\}_{i=1}^n$ are computed at the t -th epoch in Algorithm 1 of the main paper.

Output: Conditional distribution matrix $Q \in \mathbb{R}^{C \times C}$

```
1: Initialize  $Q \leftarrow \mathbf{0}^{C \times C}$  and count vector  $M \leftarrow \mathbf{0}^C$ .
2: for  $i = 1$  to  $n$  do
3:   for  $a = 1$  to  $C$  do
4:      $Q[a, y_i] \leftarrow Q[a, y_i] + \phi(F_i, a)$ 
5:   end for
6:    $M[y_i] \leftarrow M[y_i] + 1$ 
7: end for
8: for  $y = 1$  to  $C$  do
9:   for  $a = 1$  to  $C$  do
10:     $Q[a, y] \leftarrow Q[a, y]/M[y]$ 
11:   end for
12: end for
13: return  $Q$ 
```

E DIFFERENCE OF OUR HARD CHANNEL SELECTION IN DCS-ATTENTION FROM EXISTING CHANNEL ATTENTION METHODS AND ITS ADVANTAGE

The key difference of our hard channel selection in DCS-Attention from existing channel attention methods, such as SE [Hu et al., 2018], CBAM [Woo et al., 2018], MCA [Jiang et al., 2024], is that the channel selection mechanism selects an informative subset of the channels to compute the attention weights in a hard selection manner, rather than the soft reweighting scheme used in existing channel attention methods, including SE, CBAM, and MCA. Furthermore, the channels selected by DCS-Attention are token-dependent, that is, different tokens can use different channels to compute the attention weights. Such a mechanism allows for adaptive attention weights, in contrast to existing channel attention methods which use all the feature channels to compute the attention weights.

The DNNs with DCS-Attention modules under the general framework of RIB are referred to as the RIB-DCS models. The output of existing self-attention [He et al., 2021, Zhang et al., 2020] is specified by $\text{Output} = \sigma(QK^\top)V$, where $Q, K, V \in \mathbb{R}^{N \times D}$ denote the query, key, and value, respectively, with N tokens and channel dimension D . The operator $\sigma(\cdot)$ (e.g., Softmax) yields attention weights $A = \sigma(QK^\top) \in \mathbb{R}^{N \times N}$, enabling weighted aggregation over the feature tokens V . The hard channel selection mechanism in DCS-Attention uses only selected informative channels, which are the selected columns of Q and the corresponding selected rows of $K^\top$, to compute $A = \sigma(QK^\top)$, as illustrated in Fig. 3(a) of the main paper. Informative A assigns larger attention weights to semantically more relevant tokens, such as the human body regions in Fig. 1(c)–(d) of the main paper, while suppressing occlusions and background, in contrast to the existing self-attention method SPT [Tan et al., 2024].

Learning informative attention weights by our DCS-Attention through such a hard channel selection scheme can be explained with the Information Bottleneck (IB) principle. Let X denote input features, F the learned features by the DNNs, and Y the ground-truth identity labels. The IB principle aims to increase the mutual information between F and Y while decreasing that between F and X . Thus, it encourages reducing the IB loss $I(F, X) - I(F, Y)$, where $I(\cdot, \cdot)$ denotes mutual information modelling the correlation between the two input variables. By reducing the IB loss, informative attention weights are expected to reduce the correlation between F and X while increasing the correlation between F and Y . In this way, learned features F are less correlated with the input containing background or occlusions, thus mitigating the adverse effects of background or occlusions in the input, and they are more correlated with the person identity which is the goal of person Re-ID.

The channels of Q and K encode all the information in the input image, so that existing channel attention methods, such as SE, CBAM, and MCA, which use all the feature channels, inevitably suffer from the background or occlusions in the input even after a reweighting process on the feature channels, potentially hurting the performance of person Re-ID. In a strong contrast, the hard channel selection in DCS-Attention only uses a learned subset of channels to compute the attention weights, so that only a selected subset of input information is used to compute the attention weights and the correlation between the learned features F and the input X , $I(F, X)$, is decreased. Such a hard channel selection mechanism is inspired

Table 25: Comparison between RIB-DCS and ablation models that replace the channel selection mechanism in DCS-Attention with channel attention modules, including SE [Hu et al., 2018], CBAM [Woo et al., 2018], and MCA [Jiang et al., 2024]. SE (Self-Attention), CBAM (Self-Attention), and MCA (Self-Attention) denote self-attention modules that apply SE, CBAM, or MCA to reweight the channels of the query and key when computing attention weights. TransReID [He et al., 2021] is used as the backbone for all the models in this table.

Dataset	Model	FLOPs	mAP	Rank-1	IBB	$I(F, X)$	$I(F, Y)$	IB Loss
Market1501	TransReID	19.3 G	88.9	95.2	0.059	0.071	0.091	-0.020
	TransReID+SE (Self-Attention)	19.8 G	89.5	95.3	0.055	0.070	0.092	-0.023
	TransReID+CBAM (Self-Attention)	20.0 G	89.2	95.2	0.060	0.070	0.091	-0.021
	TransReID+MCA (Self-Attention)	20.3 G	89.5	95.2	0.054	0.071	0.092	-0.023
	TransReID+DCS-Attention	19.7 G	90.4	95.8	0.050	0.066	0.094	-0.028
	TransReID+IBB	19.3 G	90.2	96.0	0.040	0.062	0.105	-0.043
	RIB-SE (Self-Attention)	19.8 G	90.4	95.8	0.040	0.059	0.103	-0.044
	RIB-CBAM (Self-Attention)	20.0 G	90.1	95.7	0.041	0.063	0.102	-0.039
	RIB-MCA (Self-Attention)	20.3 G	90.3	96.2	0.035	0.060	0.108	-0.048
	RIB-DCS-FB (TransReID)	19.7 G	91.3	97.0	0.027	0.045	0.112	-0.067
DukeMTMC	TransReID	19.3 G	82.0	90.7	0.068	0.060	0.090	-0.030
	TransReID+SE (Self-Attention)	19.8 G	82.3	90.8	0.065	0.059	0.091	-0.032
	TransReID+CBAM (Self-Attention)	20.0 G	82.1	90.7	0.066	0.058	0.091	-0.033
	TransReID+MCA (Self-Attention)	20.3 G	82.3	90.6	0.067	0.060	0.092	-0.032
	TransReID+DCS-Attention	19.7 G	83.1	91.0	0.060	0.048	0.093	-0.045
	TransReID+IBB	19.3 G	83.3	91.1	0.033	0.046	0.101	-0.055
	RIB-SE (Self-Attention)	19.8 G	83.4	91.2	0.035	0.044	0.099	-0.055
	RIB-CBAM (Self-Attention)	20.0 G	83.2	91.0	0.036	0.046	0.099	-0.053
	RIB-MCA (Self-Attention)	20.3 G	83.2	91.2	0.034	0.045	0.101	-0.056
	RIB-DCS-FB (TransReID)	19.7 G	84.3	92.0	0.022	0.035	0.105	-0.070
MSMT17	TransReID	19.3 G	67.4	85.3	0.068	0.059	0.075	-0.016
	TransReID+SE (Self-Attention)	19.8 G	67.8	85.5	0.066	0.056	0.074	-0.018
	TransReID+CBAM (Self-Attention)	20.0 G	67.7	85.3	0.067	0.054	0.073	-0.019
	TransReID+MCA (Self-Attention)	20.3 G	67.5	85.5	0.069	0.058	0.074	-0.016
	TransReID+DCS-Attention	19.7 G	68.8	86.3	0.062	0.048	0.075	-0.027
	TransReID+IBB	19.3 G	68.8	86.5	0.033	0.052	0.089	-0.037
	RIB-SE (Self-Attention)	19.8 G	69.2	86.5	0.034	0.052	0.091	-0.039
	RIB-CBAM (Self-Attention)	20.0 G	68.8	86.4	0.036	0.051	0.089	-0.038
	RIB-MCA (Self-Attention)	20.3 G	68.8	86.5	0.036	0.054	0.091	-0.037
	RIB-DCS-FB (TransReID)	19.7 G	70.2	87.1	0.030	0.040	0.093	-0.053

by the IB principle. In particular, the decreased $I(F, X)$ mitigates the adverse effects of background or occlusions in the input, while maintaining a reasonable correlation between F and Y by standard training with cross-entropy loss involving Y . DCS-Attention is further enhanced by the adaptive channel selection mechanism where different tokens are allow to different channels to compute the attention weights. To demonstrate the advantage of DCS-Attention models, including TransReID+DCS-Attention and RIB-DCS-FB (TransReID), we build three ablation self-attention modules denoted as SE (Self-Attention), CBAM (Self-Attention), and MCA (Self-Attention), which perform soft channel reweighting on all the feature channels in Q and K in self-attention, using SE, CBAM, and MCA, respectively. The models that replace the self-attention modules in TransReID with SE (Self-Attention), CBAM (Self-Attention), and MCA (Self-Attention) are denoted as TransReID+SE (Self-Attention), TransReID+CBAM (Self-Attention), and TransReID+MCA (Self-Attention), respectively. The model named TransReID+DCS-Attention replaces all the self-attention modules in TransReID with DCS-Attention, which is trained with the standard cross-entropy and triplet loss without our proposed IBB to explicitly reduce the IB loss. In addition, we build three ablation RIB models referred to as RIB-SE (Self-Attention), RIB-CBAM (Self-Attention), and RIB-MCA (Self-Attention), which replace the DCS-Attention modules in RIB-DCS with the SE (Self-Attention), CBAM (Self-Attention), and MCA (Self-Attention) modules. The IBB is added to the training loss of these three ablation RIB models to explicitly reduce the IB loss. We also evaluate a baseline model, TransReID+IBB, which adds the IBB to the training loss of the vanilla TransReID without DCS-Attention.

The advantage of the hard channel selection in DCS-Attention models over the ablation self-attention models, including TransReID+SE (Self-Attention), TransReID+CBAM (Self-Attention), and TransReID+MCA (Self-Attention), is evidenced in Table 25 in Section E of the supplementary, which is also copied to this summary. In particular, Table 25 shows that the mutual information $I(F, X)$ of TransReID+DCS-Attention is smaller than that of all three ablation self-attention models, while keeping similar mutual information $I(F, Y)$ thanks to the training by the cross-entropy loss involving Y . We remark

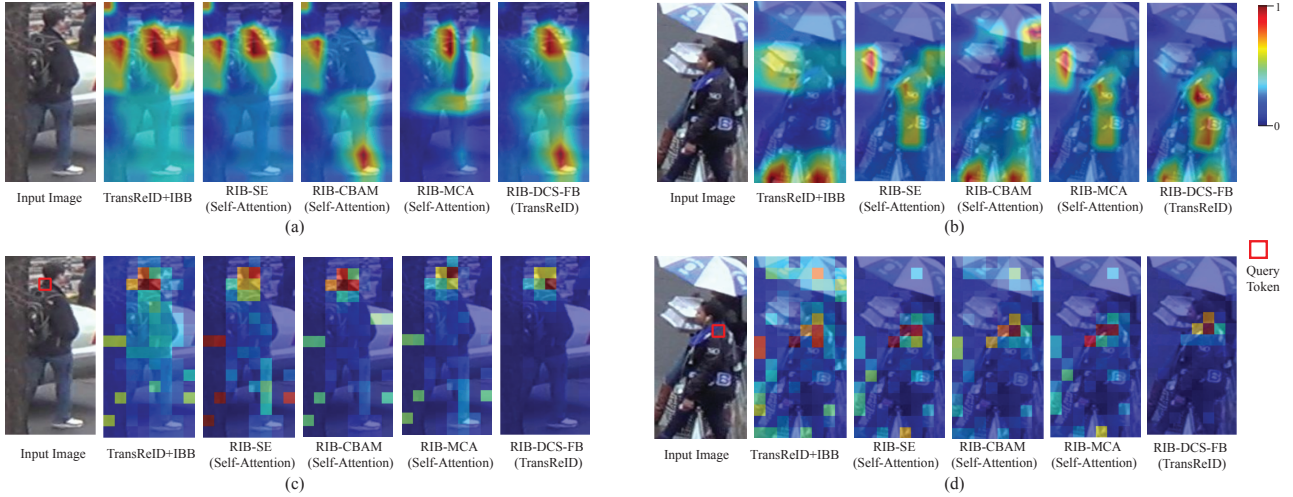


Figure 11: Figures (a)-(b) illustrate the Grad-CAM visualization for two images from DukeMTMC [Ristani et al., 2016], for TransReID+IBB, RIB-SE (Self-Attention), RIB-CBAM (Self-Attention), RIB-MCA (Self-Attention), and RIB-DCS-FB (TransReID). Figures (c)-(d) illustrate the heatmaps of the attention weights corresponding to a query token computed from the first transformer block for the above models. The query token for both examples is selected from the boundary of the neck of the person in the input images, which is critical for identifying a person.

that the IB loss can already be reduced by DCS-Attention without explicitly reducing the IB loss in the training process, mostly due to the reduced $I(F, X)$. As a result, TransReID+DCS-Attention outperforms the three ablation self-attention models in terms of both mAP and Rank-1. For example, TransReID+DCS-Attention achieves an average $I(F, X)$ of 0.054 across the three datasets in Table 25, which is 12.9% lower than that of the ablation self-attention model, TransReID+SE (Self-Attention), that is 0.062. As a result, TransReID+DCS-Attention outperforms TransReID+SE (Self-Attention) by 0.90% in average mAP and 0.83% in average Rank-1. Our RIB-DCS-FB models, termed RIB-DCS-FB (TransReID) in Table 25, further explicitly reduce the IB loss with RIB by adding the IBB to the training loss. It can be observed from Table 25 that compared to TransReID+DCS-Attention, the person Re-ID performance of RIB-DCS-FB (TransReID) is further improved, and the IB loss of RIB-DCS-FB is reduced to an even lower level, which indicates the effectiveness of the proposed RIB. For instance, RIB-DCS-FB (TransReID) reduces the average IB loss of TransReID+DCS-Attention by 0.03 on the three datasets, leading to a 1.17% improvement in average mAP and a 1.0% improvement in average Rank-1. The best of the three RIB ablation models renders only marginal improvement over that of TransReID+IBB, e.g., by 0.23% in average mAP. In contrast, RIB-DCS-FB (TransReID) outperforms TransReID+IBB by 1.17% in average mAP across the three datasets in Table 25, which demonstrates the advantages of the hard channel selection in DCS-Attention over existing channel attention methods under the general framework of RIB.

Fig. 11(c)-(d) in Section E of the supplementary, which is also copied to this summary, illustrates the heatmaps of the attention weights of TransReID+IBB, RIB-DCS (Self-Attention), RIB-SE (Self-Attention), RIB-CBAM (Self-Attention), and our RIB-DCS-FB (TransReID). With the three ablation RIB models, it is observed that even feature tokens from the occlusion and the background areas of the input image receive high attention weights, which are greater than 0.5, for a query token from the boundary of the neck of the person. For instance, all three ablation RIB models assign high attention weights, which are greater than 0.5, to regions in the middle of the tree that occlude the person in Fig. 11(c). In contrast, our proposed RIB-DCS model, RIB-DCS-FB (TransReID), assigns higher attention weights to informative and semantically relevant feature tokens, which are also located around the neck of the person in the input image. The Grad-CAM visualization in Fig. 11(a)-(b) illustrates that the three ablation RIB models either mistakenly focus on the occlusion and background areas or miss the important human body regions, potentially hurting the performance of person Re-ID. In contrast, our proposed RIB-DCS model, RIB-DCS-FB (TransReID), mostly focuses on the important human body regions in the input images. In addition, it is illustrated in Fig. 11(a)-(b) that TransReID+IBB, the model without DCS-Attention, still focuses on background or occlusion areas in the input images even with the explicit reduction of the IB loss, evidencing the necessity of DCS-Attention.

F PROOF OF THEOREM 3.1

Lemma F.1. Let $\Pr[X \in b] = \sum_{i=1}^n \mathbb{I}_{\{y_i=b\}}/n := p_b$ for every $b \in [C]$, then

$$I(F, X) \leq \frac{1}{n} \sum_{i=1}^n \sum_{a=1}^C \sum_{b=1}^C \mathbb{I}_{\{y_i=b\}} \phi(F_i, a) \log \left(\frac{\mathbb{I}_{\{y_i=b\}}}{p_b} \right). \quad (4)$$

Proof. Define $\sum_{i=1}^n \mathbb{I}_{\{y_i=b\}} := n_b$ for every $b \in [C]$, then $\Pr[X \in b] = n_b/n := p_b$. We first have

$$\begin{aligned} \Pr[F \in a, X \in b] &= \Pr[X \in b] \cdot \Pr[F \in a | X \in b] \\ &= \frac{n_b}{n} \cdot \frac{1}{n_b} \sum_{i: y_i=b} \phi(F_i, a) \\ &= \frac{1}{n} \sum_{i=1}^n \mathbb{I}_{\{y_i=b\}} \phi(F_i, a). \end{aligned} \quad (5)$$

It then follows from (5) and the log sum inequality that

$$\begin{aligned} I(F, X) &= \sum_{a=1}^C \sum_{b=1}^C \Pr[F \in a, X \in b] \log \frac{\Pr[F \in a, X \in b]}{\Pr[F \in a] \Pr[X \in b]} \\ &\leq \frac{1}{n} \sum_{i=1}^n \sum_{a=1}^C \sum_{b=1}^C \mathbb{I}_{\{y_i=b\}} \phi(F_i, a) \log (\mathbb{I}_{\{y_i=b\}} \phi(F_i, a)) - \frac{1}{n} \sum_{i=1}^n \sum_{a=1}^C \sum_{b=1}^C \mathbb{I}_{\{y_i=b\}} \phi(F_i, a) \log (\phi(F_i, a) p_b) \\ &= \frac{1}{n} \sum_{i=1}^n \sum_{a=1}^C \sum_{b=1}^C \mathbb{I}_{\{y_i=b\}} \phi(F_i, a) \log \left(\frac{\mathbb{I}_{\{y_i=b\}}}{p_b} \right). \end{aligned} \quad (6)$$

□

Lemma F.2.

$$I(F, Y) \geq \frac{1}{n} \sum_{a=1}^C \sum_{y=1}^C \sum_{i=1}^n \phi(F_i, a) \mathbb{I}_{\{y_i=y\}} \log Q(F \in a | Y = y) \quad (7)$$

Proof. Let $Q(F|Y)$ be a variational distribution. We have

$$\begin{aligned}
I(F, Y) &= \sum_{a=1}^C \sum_{y=1}^C \Pr[F \in a, Y = y] \log \frac{\Pr[F \in a, Y = y]}{\Pr[F \in a] \Pr[Y = y]} \\
&= \sum_{a=1}^C \sum_{y=1}^C \Pr[F \in a, Y = y] \log \frac{\Pr[F \in a|Y = y] Q(F \in a|Y = y)}{\Pr[F \in a] Q(F \in a|Y = y)} \\
&\geq \sum_{a=1}^C \sum_{y=1}^C \Pr[F \in a, Y = y] \log \frac{\Pr[F \in a|Y = y]}{Q(F \in a|Y = y)} + \sum_{a=1}^C \sum_{y=1}^C \Pr[F \in a, Y = y] \log \frac{Q(F \in a|Y = y)}{\Pr[F \in a]} \\
&= \text{KL}(P(F|Y) \| Q(F|Y)) + \sum_{a=1}^C \sum_{y=1}^C \Pr[F \in a, Y = y] \log \frac{Q(F \in a|Y = y)}{\Pr[F \in a]} \\
&\geq \sum_{a=1}^C \sum_{y=1}^C \Pr[F \in a, Y = y] \log \frac{Q(F \in a|Y = y)}{\Pr[F \in a]} \\
&= \sum_{a=1}^C \sum_{y=1}^C \Pr[F \in a, Y = y] \log Q(F \in a|Y = y) + H(P(F)) \\
&\geq \sum_{a=1}^C \sum_{y=1}^C \Pr[F \in a, Y = y] \log Q(F \in a|Y = y) \\
&\geq \frac{1}{n} \sum_{a=1}^C \sum_{y=1}^C \sum_{i=1}^n \phi(F_i, a) \mathbb{I}_{\{y_i=y\}} \log Q(F \in a|Y = y).
\end{aligned} \tag{8}$$

□

Proof of Theorem 3.1. Equation (1) in Theorem 3.1 of the main paper follows from $\text{IB}(\mathcal{W}) = I(F(\mathcal{W}), X) - I(F(\mathcal{W}), Y)$, the upper bound for $I(F(\mathcal{W}), X)$ in Lemma F.1 and the lower bound for $I(F(\mathcal{W}), Y)$ in Lemma F.2. □

F.1 COMPUTATION OF $Q^{(t)}(F|Y)$

The variational distribution $Q^{(t)}(F|Y)$ can be computed by

$$Q^{(t)}(F \in a|Y = y) = \Pr[F \in a|Y = y] = \frac{\sum_{i=1}^n \phi(F_i, a) \mathbb{I}_{\{y_i=y\}}}{\sum_{i=1}^n \mathbb{I}_{\{y_i=y\}}} \tag{9}$$

G CONSISTENCY OF THE MUTUAL INFORMATION ESTIMATORS IN RIB

In this section, we discuss the accuracy, or consistency, of the mutual information (MI) estimators used in the IB loss, which further justifies the soft assignment probability used in the MI computation in Section 3.1 and the link between minimizing IBB and reducing the true IB loss. The consistency of the MI estimators depends on the consistency of the estimator for the probability that the learned feature F belongs to class a , that is, $\Pr[F \in a]$, and the consistency of the estimator for the joint probability $\Pr[F \in a, X \in b]$ that the learned feature F belongs to class a and the input feature X belongs to class b . We analyze the consistency of these two probability estimators.

The estimator $\Pr[F \in a] = \frac{1}{n} \sum_{i=1}^n \phi(F_i, a)$ is a consistent plug-in estimator. Let $\mathcal{C} = \{\mathcal{C}_a\}_{a=1}^C$ denote the class centroids for the C classes, where $\mathcal{C}_a$ is the centroid of class a in the learned feature space. According to the theory of asymptotic quantization [Pollard, 1982], for a fixed C , the class centroids optimized via empirical risk minimization converge almost surely to the optimal class centroids that minimize the expected within-class quantization error,

$$\lim_{n \rightarrow \infty} \min_{\mathcal{C}} \frac{1}{n} \sum_{i=1}^n \|F_i - c_{y_i}\|_2^2 = \min_{\mathcal{C}} \mathbb{E} \left[\|F_i - c_{y_i}\|_2^2 \right]. \tag{10}$$

The class centroids in RIB are computed from the learned features according to the class labels, and thus they minimize the empirical within-class quantization objective on the left-hand side of Equation (10), guaranteeing the consistency of the learned class centroids. The asymptotic result in Equation (10) can be converted to a finite-sample result as follows. Define $R_n(\mathcal{C}) := \frac{1}{n} \sum_{i=1}^n \|F_i - c_{y_i}\|_2^2$ and $R(\mathcal{C}) := \mathbb{E} [\|F - c\|_2^2]$, and let $\hat{\mathcal{C}}_n \in \arg \min_{|\mathcal{C}|=C} R_n(\mathcal{C})$ and $\mathcal{C}_n^* \in \arg \min_{|\mathcal{C}|=C} R(\mathcal{C})$. Then, under standard regularity conditions, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$, we have

$$R(\hat{\mathcal{C}}_n) - R(\mathcal{C}_n^*) \leq \varepsilon_{n,\delta}, \quad \text{where} \quad \varepsilon_{n,\delta} = O\left(\sqrt{\frac{1}{n} \log \frac{1}{\delta}}\right). \quad (11)$$

This shows that the empirical class centroids used in RIB are consistent finite-sample approximations to the population class centroids. Furthermore, under the softmax formulation, the soft assignment $\phi(F_i, a)$ acts as a continuous relaxation of the hard assignment. The consistency of the estimator $\Pr[F \in a]$ then follows from the standard kernel density estimation literature [Einmahl and Mason, 2005, Goldstein and Messer, 1992, Rigollet and Vert, 2009]. Therefore, $\Pr[F \in a]$ is a consistent plug-in estimator of the probability that the learned feature F belongs to class a .

The joint probability estimators are consistent. Following the same argument, the estimator for the joint probability $\Pr[F \in a, Y = y] = \frac{1}{n} \sum_{i=1}^n \mathbb{I}_{\{y_i=y\}} \phi(F_i, a)$ is also consistent, since it is the empirical average of a bounded and consistent soft-assignment function $\phi(F_i, a)$ multiplied by the class indicator $\mathbb{I}_{\{y_i=y\}}$. For the MI term $I(F, X)$, RIB further uses class centroids in the input-feature space. Following the same class-centroid consistency argument, the input-feature class centroids consistently estimate their population counterparts, and thus the soft assignment $\phi(X_i, b)$ is also consistent. Consequently, the joint estimator $\Pr[F \in a, X \in b] = \frac{1}{n} \sum_{i=1}^n \phi(F_i, a) \phi(X_i, b)$ is a consistent plug-in estimator for the joint probability that F belongs to class a and X belongs to class b .

Consistency of the MI estimators and the link between IBB and the IB loss. The consistency of the estimators $\Pr[F \in a]$, $\Pr[F \in a, Y = y]$, and $\Pr[F \in a, X \in b]$ guarantees the consistency of the corresponding plug-in estimators for the two MI terms, $I(F, Y)$ and $I(F, X)$. Therefore, the MI estimation used in RIB is consistent under standard regularity conditions. Because of the consistency of the aforementioned estimators for the MI terms in Section 3.1, and the fact that our variational upper bound IBB is the upper bound for the IB loss computed using these MI estimators, IBB is provably an upper bound for the true IB loss with consistency. As a result, reducing IBB reduces the true IB loss.

H LIMITATIONS

A potential limitation of RIB is that, when using class-centroid soft assignment, the number of centroids is fixed to the number of classes C . This limitation can be addressed by using prototype/clustering centroids instead of class centroids, as described in the extension of RIB to prototypical learning (RIBPL) in Section A.18. As shown in Tables 19 and 20, RIBPL further improves the modeling capability over that of class centroids by using a larger number of prototypes, and its performance remains stable with respect to the number of prototypes. In addition, RIB currently focuses on person Re-ID with attention-based DNNs. As an independent contribution, the RIB framework is applicable to broader discriminative tasks beyond person Re-ID, and applying RIB to other discriminative tasks and modalities is left as future work.